

Bimetallic Au_xFe_y ($x + y = 6$) clusters as catalysts in the oxidation of methane, by hydrogen peroxide, for the CH_3OH synthesis

(Supporting Information)

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1. Methodological Validation

1.1. Organic molecules

The GS geometries of the studied molecules were optimized at the PBE0/Def2TZVP level. Average equilibrium bond lengths are summarized in Table S1. The calculated values are in good agreement with the experimental data reported by NIST, with deviations below 0.02 Å for all bond distances. This agreement validates the chosen method for describing both saturated and radical species.

Table S1. Average bond lengths (Å) for the CH₄, H₂O₂, •CH₃, and •OH species.

Molecule		Bond	Theoretical (PBE0/Def2TZVP)	Experimental (NIST)
CH ₄	C–H		1.090	1.087(7)
•CH ₃	C–H		1.080	1.088(1)
H ₂ O ₂	O–H		0.965	0.950(5)
•OH	O–H		0.973	0.970(9)
CH ₃ OH	O–H		0.959	0.956(15)
H ₂ O	O–H		0.959	0.9572(9)
CH ₃ OH	C–O		1.409	1.427(7)
H ₂ O ₂	O–O		1.426	1.475(4)

Dipole moments (**Table S2**) are also reproduced accurately, further supporting the reliability of the determined structural and electronic properties.

Table S2. Dipole moments (D) for the CH₄, H₂O₂, •CH₃, and •OH species.

Molecule	Theoretical	Experimental (NIST)
CH ₄	0.000	0.000
H ₂ O ₂	1.861	1.770
•CH ₃	0.000	0.000
•OH	1.769	1.668
CH ₃ OH	1.674	1.672
H ₂ O	2.068	1.857

Vertical ionization potentials were estimated from single-point calculations for neutrals and cations. Theoretical results (**Table S3**) show good agreement with the experiment. Maximum deviation, of ~0.48 eV, appears for methane.

Table S3. Vertical ionization potentials (eV) for CH₄, H₂O₂, •CH₃, •OH, •H, CH₃OH, and H₂O.

Molecule	Theoretical	Experimental (NIST)
CH ₄	14.079	13.600
H ₂ O ₂	11.427	11.700
•CH ₃	9.958	9.840
•OH	13.003	13.017
•H	13.634	13.610
CH ₃ OH	10.781	10.960
H ₂ O	12.525	12.500

In contrast, vertical electron affinities (VEAs), found at the same level (Table S4), deviate significantly from literature values, often overestimating anion stabilities by >2 eV. This limitation is attributed to (i) the absence of diffuse functions in Def2TZVP and (ii) the self-interaction error of hybrid functionals, which overestimates LUMO energies. These results highlight the need for augmented basis sets and higher-level methods when accurate electron affinities are required.

Table S4. Vertical electron affinities (eV) for CH₄, H₂O₂, •CH₃, •OH, •H, CH₃OH, and H₂O.

Molecule	Theoretical	Reported
CH ₄	2.457	0.5–1.2
H ₂ O ₂	2.206	0.0434
•CH ₃	0.537	0.080 ± 0.030
•OH	2.252	1.828
•H	0.037	0.754
CH ₃ OH	2.103	0.893
H ₂ O	2.069	0.800

Chemical potentials, found from the VEAs and VIPs, are summarized in **Table S5**. Deviations with respect to experimental data are <1 eV, supporting the qualitative trends used to discuss reactivity.

Table S5. Chemical potentials (μ , eV) for $\bullet\text{CH}_3$, CH_4 , H_2O_2 , H_2O , $\bullet\text{H}$, and $\bullet\text{OH}$.

Molecule	μ_{theo}	μ_{exp}
$\bullet\text{CH}_3$	-5.248	-4.960
CH_4	-8.268	-7.400
H_2O_2	-6.816	-5.872
H_2O	-7.297	-6.700
$\bullet\text{H}$	-6.835	-7.182
$\bullet\text{OH}$	-7.642	-7.423

Finally, calculated vibrational frequencies (**Table S6**) show good agreement with experimental data, supporting the reliability of the optimized geometries.

Table S6. Vibrational frequencies (cm^{-1}) for CH_4 , and H_2O_2 .

Molecule	Mode	Theoretical	Experimental
CH_4	$\nu(\text{C-H})$	3047	3009
H_2O_2	$\nu(\text{O-O})$	1007	845–875

1.2. Bare clusters

Here, the GSs of Au_xFe_y , $x + y = 6$, were determined by generating randomized atomic coordinates as well as by considering previous reported geometries [35,36]. Random atomic geometry configurations of bare Au_xFe_y clusters ($x + y = 6$) were generated using an in-house Python script [52]. Structural optimization was performed by exploring these chosen candidates, leading to the identification of ground and excited states, (Figure S1). Both two-dimensional (2D) and three-dimensional (3D) motifs were observed within a narrow energy range. Thus, Au–Au bonds are the longest and Fe–Fe the shortest, yielding more compact structures as the iron content increases (Table S7); which is accounted from the fact that the Fe atoms form multiple 3d–3d and 4s–4s bonding while the Au atoms mainly forms 6s–6s bond patterns. This bond distance trend is relevant for catalytic properties [53], together with the interplay of Au–Fe interactions, since weak heterometallic bonding has been associated with the observed enhanced activity [13].

The GS of **Au₆** corresponds to a slightly distorted planar triangular geometry (Figure S1). Due to its triangular arrangement, two types of Au atoms are present with distinct bond lengths (Table S7). These findings are consistent with the triangular geometries reported for neutral gold clusters by Walker, Assadollahzadeh, and Schwerdtfeger [54,55]. Walker also found rhombus and capped bipyramid motifs as excited states. Here, those structures appear with higher spin multiplicities (Figure S1). Boronat et al. [8] experimentally confirmed that small planar Au clusters are more active in redox processes than 3D nanoparticles of ~1 nm, following radical pathways involving peroxo or hydroperoxo intermediates.

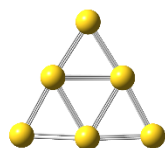
For **Au₅Fe**, the GS exhibits a chevron-type geometry (C_{2v}) with the Fe atom located at the center and three types of Au atoms bonded to it (Figure S1). Excited states include a distorted triangle and a 3D geometry. These structures are consistent with geometries reported by Mawale et al. [56] for Au₅Fe⁺ charged clusters. In their study, they synthesized bimetallic nanoparticles and performed DFT optimizations using the ZORA code, with a scalar relativistic approximation, to explore the structures of Au₅Fe, Au₇Fe₂, and Au₉Fe₂, as representative examples of larger assemblies.

We found that the GS geometries of **Au₄Fe₂** and **Au₃Fe₃** (C_1) are markedly asymmetric (Figure S1). Au₄Fe₂ features three types of Au atoms and two Fe sites (Figure S1). Excited states include both 2D and 3D motifs. These structures are consistent with those reported by Woldegebriel [57] who employed DFT *ab-initio* calculations using the PW91 functional for exchange–correlation energy, and the projector augmented wave (PAW) method.

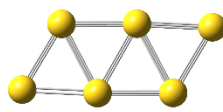
Au₃Fe₃ shows three types of Au atoms and three Fe sites (Figure S1). In Au₃Fe₃ the Fe atoms suffer segregation, forming a Fe₃ triangle where the Au atoms bond surrounding the Fe sites. Such segregation persists as the content of iron atoms increases. The 3D GS geometry found here for Au₃Fe₃ differs from the 2D GS structure reported by Woldegebriel [57], where a Au₃Fe₃ planar triangle is formed. Suggesting that the GS of Au₃Fe₃ may depend sensitively on the chosen level of theory; functionals as well as basis sets. Further theoretical and experimental studies are needed to clarify the most stable states of Au_xFe_y clusters.

The GS's of **Au₂Fe₄** and **AuFe₅** (C_2) exhibit low symmetry (Figure S1). Au₂Fe₄ shows a bi-capped octahedron with two types of Au atoms and two Fe sites. The GS of AuFe₅ displays six different atomic sites. Its excited states are energetically near to the GS. These results are consistent with geometries reported by Woldegebriel [57].

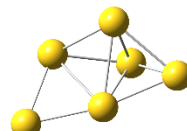
Fe₆ clusters adopt 3D geometries (Figure S1). The GS corresponds to a distorted bipyramid (D_{3h}) with high spin, featuring three types of Fe atoms and two Fe–Fe distances (Table S7). This agrees with previous reports [35, 56]. Excited states are higher spin structures (Figure S1).



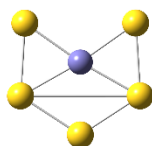
Au₆ M1
GS, D_{3h}



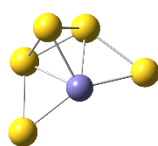
Au₆ M3
1.459 eV, C_{2h}



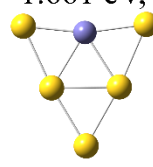
Au₆ M3
1.661 eV, C₁



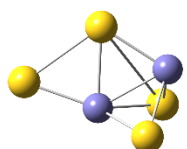
Au₅Fe M4
GS, C_{2v}



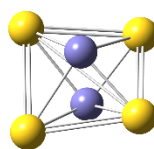
Au₅Fe M4
0.569 eV, C₁



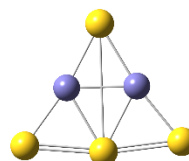
Au₅Fe M6
0.583 eV, C_{2v}



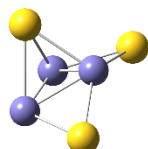
Au₄Fe₂ M9
GS, C₁



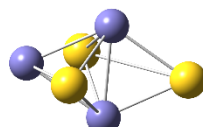
Au₄Fe₂ M9
0.425 eV, C₂



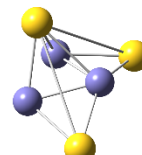
Au₄Fe₂ M7
0.497 eV, C_s



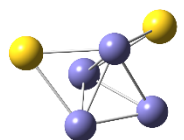
Au₃Fe₃ M12
GS, C₁



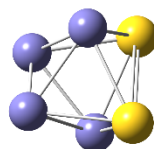
Au₃Fe₃ M12
0.130 eV, C_s



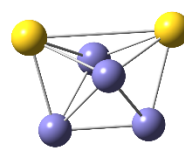
Au₃Fe₃ M10
0.425 eV, C₁



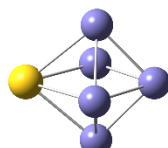
Au₂Fe₄ M15
GS, C₂



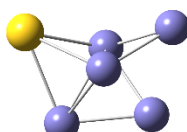
Au₂Fe₄ M15
0.226 eV, C₁



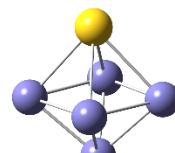
Au₂Fe₄ M17
0.540 eV, C_s



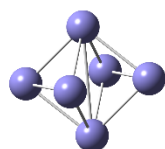
AuFe₅ M18
GS, C₂



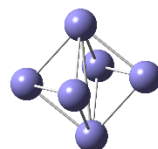
AuFe₅ M18
0.014 eV, C₁



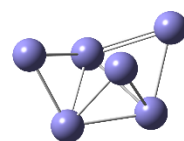
AuFe₅ M18
0.028 eV, C₁



Fe₆ M21
GS, D_{4h}



Fe₆ M23
0.609 eV, C_{2h}



Fe₆ M23
0.701 eV, C₁

Figure S1. Optimized ground and excited state geometries of Au_xFe_y clusters ($x + y = 6$) computed at the PBE0/ Def2TZVP level with def2-TZVP (Fe) and LANL2DZ (Au). Au-rich clusters are planar or slightly distorted, while Fe-rich clusters adopt compact three-dimensional, lower-symmetry structures. Spin multiplicities are indicated.

Table S7. Average bond lengths between atoms per species.

Species	$\bar{d}_{\text{Au-Au}}$	$\bar{d}_{\text{Fe-Fe}}$	$\bar{d}_{\text{Au-Fe}}$	$\bar{d}_{\text{Au-C}}$	$\bar{d}_{\text{Au-O}}$	$\bar{d}_{\text{Fe-C}}$	$\bar{d}_{\text{Fe-O}}$	$\bar{d}_{\text{O-C}}$	$\bar{d}_{\text{O-O}}$	$\bar{d}_{\text{C-H}}$	$\bar{d}_{\text{O-H}}$
CH ₄										1.090	
H ₂ O ₂									1.426		0.965
H ₃ C•										1.080	
HO•											0.973
H•											
CH ₃ OH								1.409		1.095	0.959
H ₂ O											0.959
Bare clusters											
Au ₆	2.725										
Au ₅ Fe	2.725		2.610								
Au ₄ Fe ₂	3.081	2.584	2.574								
Au ₃ Fe ₃		2.482	2.615								
Au ₂ Fe ₄		2.483	2.650								
AuFe ₅		2.458	2.689								
Fe ₆		2.466									
CH ₄ adsorption											
Au ₆ CH ₄	2.724			3.065						1.092	
Au ₅ FeCH ₄	2.723		2.610	3.188						1.091	
Au ₄ Fe ₂ CH ₄	3.018	2.601	2.562			3.153				1.092	
Au ₃ Fe ₃ CH ₄		2.483	2.621			2.651				1.094	
Au ₂ Fe ₄ CH ₄		2.489	2.657			2.508				1.095	
AuFe ₅ CH ₄		2.459	2.697			2.657				1.094	
Fe ₆ CH ₄		2.474				3.014				1.092	
H ₂ O ₂ adsorption											
Au ₆ H ₂ O ₂	2.724				2.501				1.425		0.967
Au ₅ FeH ₂ O ₂	2.714		2.648				2.146		1.424		0.975
Au ₄ Fe ₂ H ₂ O ₂	2.967	2.634	2.561				2.156		1.426		0.976
Au ₃ Fe ₃ H ₂ O ₂		2.502	2.622				2.155		1.426		0.978
Au ₂ Fe ₄ H ₂ O ₂		2.491	2.669				2.211		1.426		0.976
AuFe ₅ H ₂ O ₂		2.461	2.709				2.185		1.430		0.979
Fe ₆ H ₂ O ₂		2.479					2.219		1.430		0.980
Methyl radical Formation											
Au ₆ H ₃ C•	2.735			2.063						1.090	
Au ₅ FeH ₃ C•	2.723		2.653	2.055						1.091	
Au ₄ Fe ₂ H ₃ C•	2.738	2.681	2.593			1.996				1.092	
Au ₃ Fe ₃ H ₃ C•		2.431	2.627			2.165				1.100	
Au ₂ Fe ₄ H ₃ C•		2.585	2.616			2.020				1.096	
AuFe ₅ H ₃ C•		2.471	2.648			2.148				1.101	
Fe ₆ H ₃ C•		2.521				2.028				1.097	
Hydroxyl radical Formation											
Au ₆ HO•	2.725				1.986						0.964
Au ₅ FeHO•	2.738		2.684				1.794				0.957
Au ₄ Fe ₂ HO•	3.010	2.614	2.638				1.797				0.958
Au ₃ Fe ₃ HO•		2.471	2.623				1.956				0.958
Au ₂ Fe ₄ HO•		2.510	2.638				1.804				0.985
AuFe ₅ HO•		2.532	2.713				1.811				0.956

Fe ₆ HO•			2.514				1.813				0.956
Transition States											
(Au ₆ OH—CH ₃) _{ts}	2.732			2.248	2.111			2.027		1.085	0.965
(Au ₅ FeOH—CH ₃) _{ts}	2.736		2.688				1.851	2.115		1.080	0.962
(Au ₄ Fe ₂ OH—CH ₃) _{ts}	2.980	2.613	2.630				1.872	2.071		1.079	0.964
(Au ₃ Fe ₃ OH—CH ₃) _{ts}		2.470	2.632				2.089	1.891		1.081	0.974
(Au ₂ Fe ₄ OH—CH ₃) _{ts}		2.504	2.659				1.891	2.042		1.080	0.965
(AuFe ₅ OH—CH ₃) _{ts}		2.498	2.714				1.881	2.012		1.081	0.959
(Fe ₆ OH—CH ₃) _{ts}		2.476					1.861	2.202		1.079	0.963
Cluster – Methanol											
Au ₆ —CH ₃ OH	2.725				2.411			1.425		1.091	0.961
Au ₅ Fe—CH ₃ OH	2.713		2.651				2.123	1.427		1.091	0.964
Au ₄ Fe ₂ —CH ₃ OH		2.663	2.595				2.097	1.433		1.090	0.966
Au ₃ Fe ₃ —CH ₃ OH		2.491	2.632				2.134	1.432		1.090	0.962
Au ₂ Fe ₄ —CH ₃ OH		2.495	2.664				2.168	1.427		1.091	0.967
AuFe ₅ —CH ₃ OH		2.463	2.707				2.133	1.430		1.090	0.962
Fe ₆ —CH ₃ OH		2.481					2.142	1.430		1.091	0.963

1.2.1. Magnetic Properties

Because methane oxidation proceeds through radical intermediates, the magnetic properties of the Au_xFe_y clusters may play an important role in the activation and stabilization of open-shell species. As the proportion of iron atoms in Au_xFe_y increases, the number of unpaired electrons also rises, with the total spin progressing from S = 0 for pure gold Au₆ to S = 20 for pure iron Fe₆ cluster (Table S8). In all mixed clusters, the largest magnetic moments are primarily localized on Fe atoms, whereas Au atoms exhibit only small, induced spin densities. This progressive spin polarization directly influences the activation and stabilization of radical intermediates. In particular, Fe-rich compositions provide open-shell environments that promote spin transfer and facilitate radical coupling by reducing spin constraints along the reaction pathway. This magnetic character is expected to play a crucial role in the activation of radical intermediates, particularly by stabilizing short-lived radical species and facilitating key steps of the reaction. Magnetic catalysis may involve a transference of electrons with an associated transference of spin, which can be seen from the changes in the spin state in going from neutral to charged clusters [58].

Table S8. Magnetic moments and spin contamination of all the species.

Species	M (2S+1)	$\bar{\mu}_{Au}$ (μ_B)	$\bar{\mu}_{Fe}$ (μ_B)	μ_C (μ_B)	μ_O (μ_B)	$\bar{\mu}_H$ (μ_B)	spin contam ination (%)
CH ₄	1			0.000		0.000	0.00
H ₂ O ₂	1				0.000	0.000	0.00
H ₃ C•	2			1.134		-0.045	0.49

HO•	2				1.025	-0.025	0.33
H•	2					1.000	0.00
CH ₃ OH	1			0.000	0.000	0.000	0.00
H ₂ O	1				0.000	0.000	0.00
Bare clusters							
Au ₆	1	0.000					0.00
Au ₅ Fe	4	-0.112	3.561				0.34
Au ₄ Fe ₂	9	0.107	3.786				0.34
Au ₃ Fe ₃	12	0.079	3.587				0.59
Au ₂ Fe ₄	15	0.028	3.486				0.54
AuFe ₅	18	0.105	3.379				0.53
Fe ₆	21		3.333				0.65
CH ₄ adsorption							
Au ₆ CH ₄	1	0.000		0.000		0.000	0.00
Au ₅ FeCH ₄	4	-0.111	3.557	-0.003		0.001	0.35
Au ₄ Fe ₂ CH ₄	9	0.123	3.752	0.000		0.001	0.31
Au ₃ Fe ₃ CH ₄	12	0.052	3.291	-0.045		0.005	0.58
Au ₂ Fe ₄ CH ₄	15	0.056	3.476	-0.015		0.000	0.50
AuFe ₅ CH ₄	18	0.158	3.373	-0.025		0.001	0.51
Fe ₆ CH ₄	21		3.335	-0.017		0.001	0.69
H ₂ O ₂ adsorption							
Au ₆ H ₂ O ₂	1	0.000			0.000	0.000	0.00
Au ₅ FeH ₂ O ₂	4	-0.118	3.566		0.007	0.003	0.45
Au ₄ Fe ₂ H ₂ O ₂	9	0.138	3.713		0.005	0.005	0.28
Au ₃ Fe ₃ H ₂ O ₂	12	0.096	3.561		0.003	0.012	0.65
Au ₂ Fe ₄ H ₂ O ₂	15	0.066	3.467		-0.001	0.002	0.48
AuFe ₅ H ₂ O ₂	18	0.162	3.369		0.001	-0.007	0.52
Fe ₆ H ₂ O ₂	21		3.339		0.002	-0.019	0.73
Methyl radical Formation							
Au ₆ H ₃ C•	2	0.100		0.431		0.000	1.17
Au ₅ FeH ₃ C•	5	0.039	3.745	0.062		0.000	0.43
Au ₄ Fe ₂ H ₃ C•	8	-0.046	3.633	-0.110		0.009	2.67
Au ₃ Fe ₃ H ₃ C•	11	0.052	3.291	-0.045		0.005	0.77
Au ₂ Fe ₄ H ₃ C•	16	0.143	3.684	-0.042		0.007	0.35
AuFe ₅ H ₃ C•	17	-0.090	3.237	-0.104		0.003	0.85
Fe ₆ H ₃ C•	22		3.504	-0.039		0.006	0.56
Hydroxyl radical Formation							
Au ₆ HO•	2	0.114			0.323	-0.008	1.39
Au ₅ FeHO•	5	0.025	3.682		0.188	0.007	0.97
Au ₄ Fe ₂ HO•	8	0.067	3.290		0.145	0.005	0.50
Au ₃ Fe ₃ HO•	11	0.024	3.279		0.087	0.004	1.02
Au ₂ Fe ₄ HO•	14	0.041	3.198		0.125	0.000	1.08
AuFe ₅ HO•	19	0.287	3.517		0.122	0.005	0.33
Fe ₆ HO•	22		3.480		0.122	0.005	0.54
Transition States							
(Au ₆ OH—CH ₃) _{ts}	1	0.000		0.000	0.000	0.000	0.00
(Au ₅ FeOH—CH ₃) _{ts}	4	-0.002	3.553	-0.719	0.096	0.020	1.21
(Au ₄ Fe ₂ OH—CH ₃) _{ts}	9	-0.005	3.656	0.742	0.011	-0.011	1.83
(Au ₃ Fe ₃ OH—CH ₃) _{ts}	12	0.056	3.440	0.586	-0.063	-0.003	0.69
(Au ₂ Fe ₄ OH—CH ₃) _{ts}	15	0.046	3.311	0.744	-0.035	-0.007	0.69
(AuFe ₅ OH—CH ₃) _{ts}	18	0.269	3.451	-0.667	0.077	0.016	0.54
(Fe ₆ OH—CH ₃) _{ts}	21		3.217	0.841	-0.064	-0.020	0.78

Cluster – Methanol							
Au ₆ –CH ₃ OH	1	0.000		0.000	0.000	0.000	0.00
Au ₅ Fe–CH ₃ OH	4	-0.122	3.586	0.000	0.014	0.002	0.46
Au ₄ Fe ₂ –CH ₃ OH	9	0.149	3.696	0.002	0.007	0.001	0.22
Au ₃ Fe ₃ –CH ₃ OH	12	0.106	3.560	-0.002	-0.003	0.002	0.59
Au ₂ Fe ₄ –CH ₃ OH	15	0.070	3.463	-0.002	-0.011	0.006	0.56
AuFe ₅ –CH ₃ OH	18	0.180	3.366	-0.003	-0.008	0.000	0.50
Fe ₆ –CH ₃ OH	21		3.336	-0.006	-0.014	0.002	0.71

In contrast, the absence of magnetic moments in the closed-shell Au₆ motif suggests limited or selective reactivity toward radical species via spin-polarized or charge-transfer pathways. However, adsorption of neutral CH₄ and H₂O₂ molecules remains possible, likely driven by weak electrostatic interactions and lone-pair donation from the oxygen atoms to the positively charged Au centers. Notably, Au₅Fe features a high-spin iron center, whereas the gold atoms present small induced magnetic moments, marking the appearance of the first open-shell system within the series. All bimetallic clusters exhibit significant magnetic moments localized mainly on the iron atoms (Table S8), while smaller induced spin densities appear on gold sites.

The high-spin character is most pronounced in Fe-rich systems, indicating that iron incorporation enhances the magnetic response of the Au_xFe_y clusters with respect to pure gold motifs. Overall, this picture of open-shell configurations in the Au_xFe_y particles highlights their potential for high reactivity toward neutral and radical species via spin-polarized and charge-transfer mechanisms. In particular, the Fe sites, which concentrate the largest magnetic moments, are expected to play a central role in the adsorption and activation of the molecules, as well as in the reduction of the radical formation barriers discussed below. Detailed local magnetic moments are summarized in Table S8.

Beyond magnetism, the catalytic behavior of Au_xFe_y clusters is closely linked to their electronic structure, which governs charge redistribution and electron transfer during adsorption and radical formation steps.

1.2.2. Energetics and reactivity

Because methane oxidation involves adsorption, charge redistribution, and radical formation, the electronic structure of Au_xFe_y clusters plays a key role in their catalytic behavior. In particular, properties such as vertical ionization potentials (VIPs) and electron affinities (VEAs), or charge

distribution provide insight into their ability to donate and accept electron density during bond activation and the stabilization of radical intermediates.

Table S9. Electronic reactivity descriptors of Au_xFe_y clusters

Cluster	Multiplicity	VIP (eV)	VEA (eV)	HOMO–LUMO gap (eV)
Au_6	1	8.287	-1.421	9.873
Au_5Fe	4	8.714	-1.489	10.004
Au_4Fe_2	9	7.598	-1.106	8.812
Au_3Fe_3	12	6.348	-1.153	7.448
Au_2Fe_4	15	5.915	-0.723	6.930
AuFe_5	18	6.417	-0.588	7.043
Fe_6	21	6.418	-0.807	6.740

As the Fe content increases, the VIPs and HOMO–LUMO gaps generally decrease (Figure S2 and Table S9), reflecting progressively more reactive electronic structures. Among the studied systems, Au_2Fe_4 exhibits the lowest VIP, indicating a strong propensity for electron donation, whereas Au-rich clusters retain larger gaps and higher ionization potentials. Overall, these trends suggest that Fe incorporation enhances charge redistribution within the clusters.

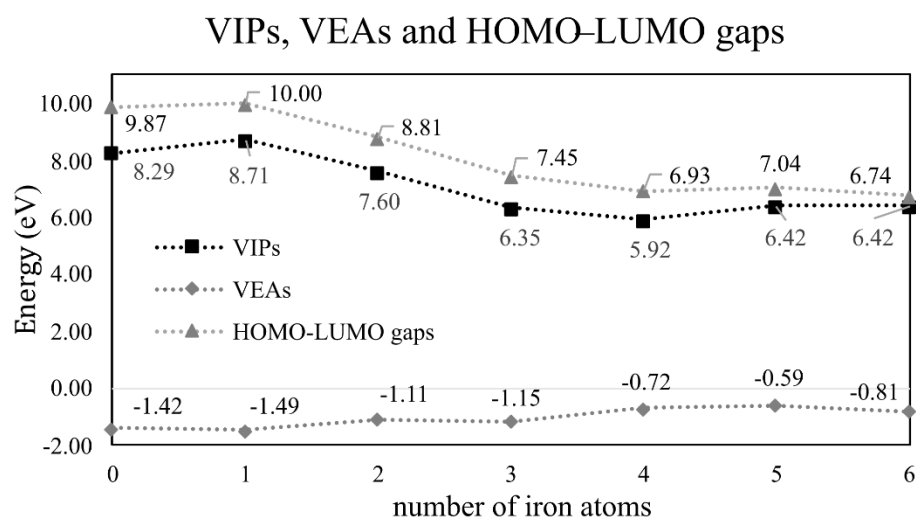


Figure S2. Evolution of VIPs, VEAs, and HOMO–LUMO gaps of Au_xFe_y clusters ($x + y = 6$) as a function of Fe content, showing a general decrease in ionization potentials and energy gaps with increasing Fe concentration.

2. Obtention of radicals

2.1. Energetics

Consistent with the estimated bond lengths and vibrational frequencies (Table S10), radical formation becomes energetically more favorable upon CH₄ and H₂O₂ adsorption on Au_xFe_y. So, the CH₄ → •CH₃ + •H process requires 4.29 eV in the gas phase, but this barrier decreases to 2.14 – 3.50 eV on the Au_xFe_y-CH₄ complexes. The lowest free energy was observed in Au₃Fe₃ (Figure S3). In the case of hydrogen peroxide, the H₂O₂ → 2•OH event, which presents a free energy of 1.69 eV in the gas phase, becomes spontaneous in most cluster–adsorbate systems.

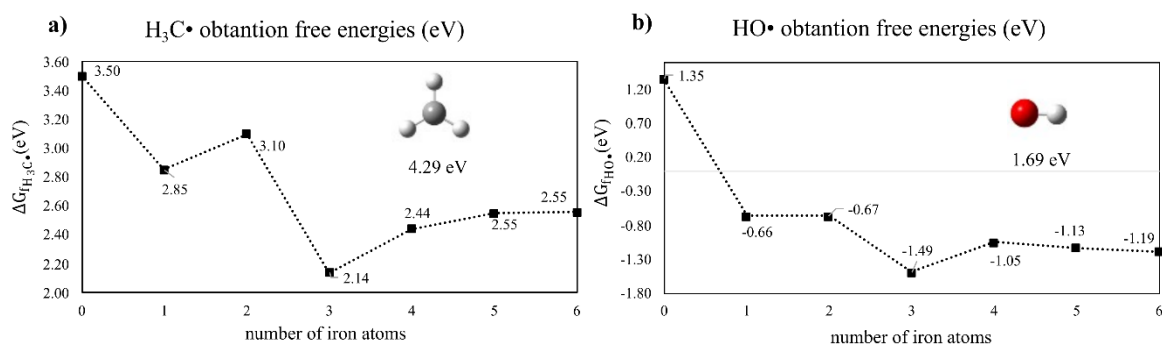


Figure S3. Determination of free energies for a) H₃C• and b) HO• radicals complexes, as the number of iron atoms.

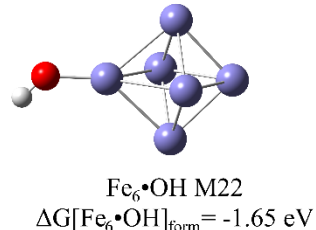
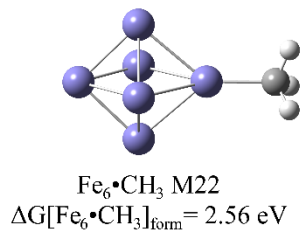
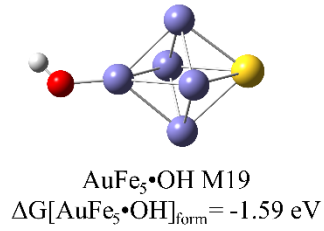
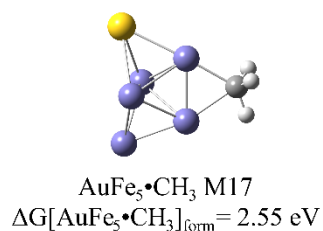
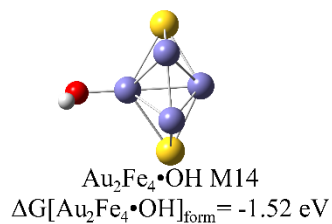
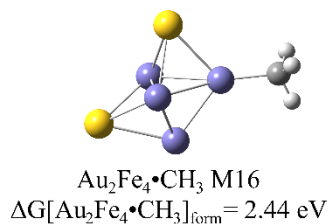
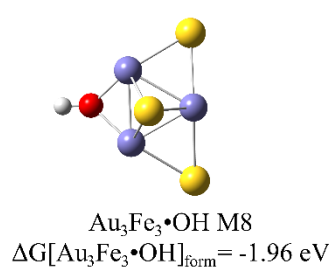
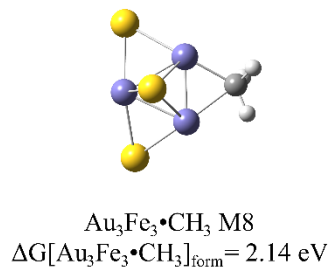
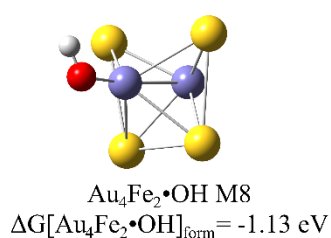
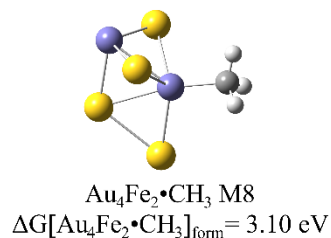
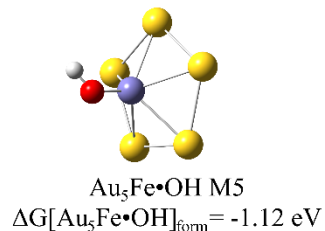
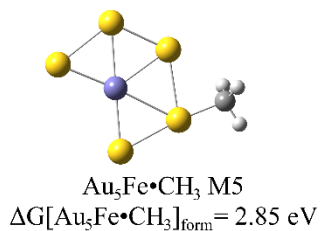
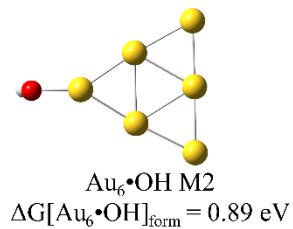
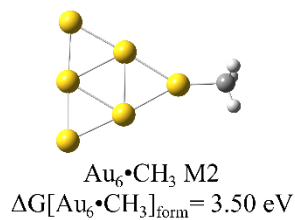


Figure S4. $\text{Au}_x\text{Fe}_y\text{CH}_4 \rightarrow \text{Au}_x\text{Fe}_y\bullet\text{CH}_3 + \bullet\text{H}$, and $\text{Au}_x\text{Fe}_y\text{H}_2\text{O}_2 \rightarrow \text{Au}_x\text{Fe}_y\text{HO}\bullet + \bullet\text{OH}$ formation free energies and radical complexes multiplicities, where $x + y = 6$.

The only exception is the adsorption on the pure gold cluster, Au_6 , where the process still needs 0.89 eV, while the most favorable case is Au_3Fe_3 , reaching -1.96 eV (Figure S4). Once the radicals are formed, structural rearrangements are evident: cluster–C and cluster–O bond distances become considerably shorter compared to those in the reactant complexes (Table S7), suggesting stronger metal–radical interactions. These structural changes are also reflected in the vibrational frequencies, confirming the more robust bonding between radicals and the metallic clusters, Table S10. Note that the less shorter Fe–C and Fe–O bond lengths as well as the smaller Fe–C and Fe–O vibrational frequencies takes place on the corresponding $\text{Au}_2\text{Fe}_3\bullet\text{CH}_3$ and $\text{Au}_3\text{Fe}_3\bullet\text{CH}_3$ complexes, indicating that they may have the suitable reactivity for the occurrence of the further reaction steps. Therefore, Au_3Fe_3 is energetically feasible for the $\bullet\text{CH}_3$ or $\bullet\text{OH}$ radical formation (Figure S5).

These properties provide further evidence of bond activation, induced by the cluster interactions. After radical formation, the overall trends are mostly preserved; however, some distinctions arise. The highest ionization potential was found for $\text{Au}_6\bullet\text{OH}$ (8.144 eV), while the lowest corresponded to $\text{Au}_2\text{Fe}_4\bullet\text{CH}_3$ (5.800 eV). In terms of electron affinity, $\text{Au}_6\bullet\text{OH}$ exhibited the most negative value (-2.798 eV), whereas $\text{Au}_2\text{Fe}_4\bullet\text{OH}$ showed the least negative one (-0.475 eV). When methane and hydrogen peroxide are adsorbed, a net charge transfer occurs from the cluster toward the atoms directly involved in the bonding (Table S11). This redistribution of electronic density is reflected in the partial charges of the bonded atoms, providing evidence of bond activation and the strengthening of cluster–adsorbate bond lengths. Considering these values, along with the systematic decrease of the HOMO–LUMO gaps as the proportion of Fe atoms increases, clusters containing three or more iron atoms are expected to display higher reactivity. Once the radicals are formed, the unbalance of the cluster–radical charge systems increases significantly (Table S11), a trend that is in line with the redistribution of charge observed on the bonded C and O atoms (q_C , and Q_O).

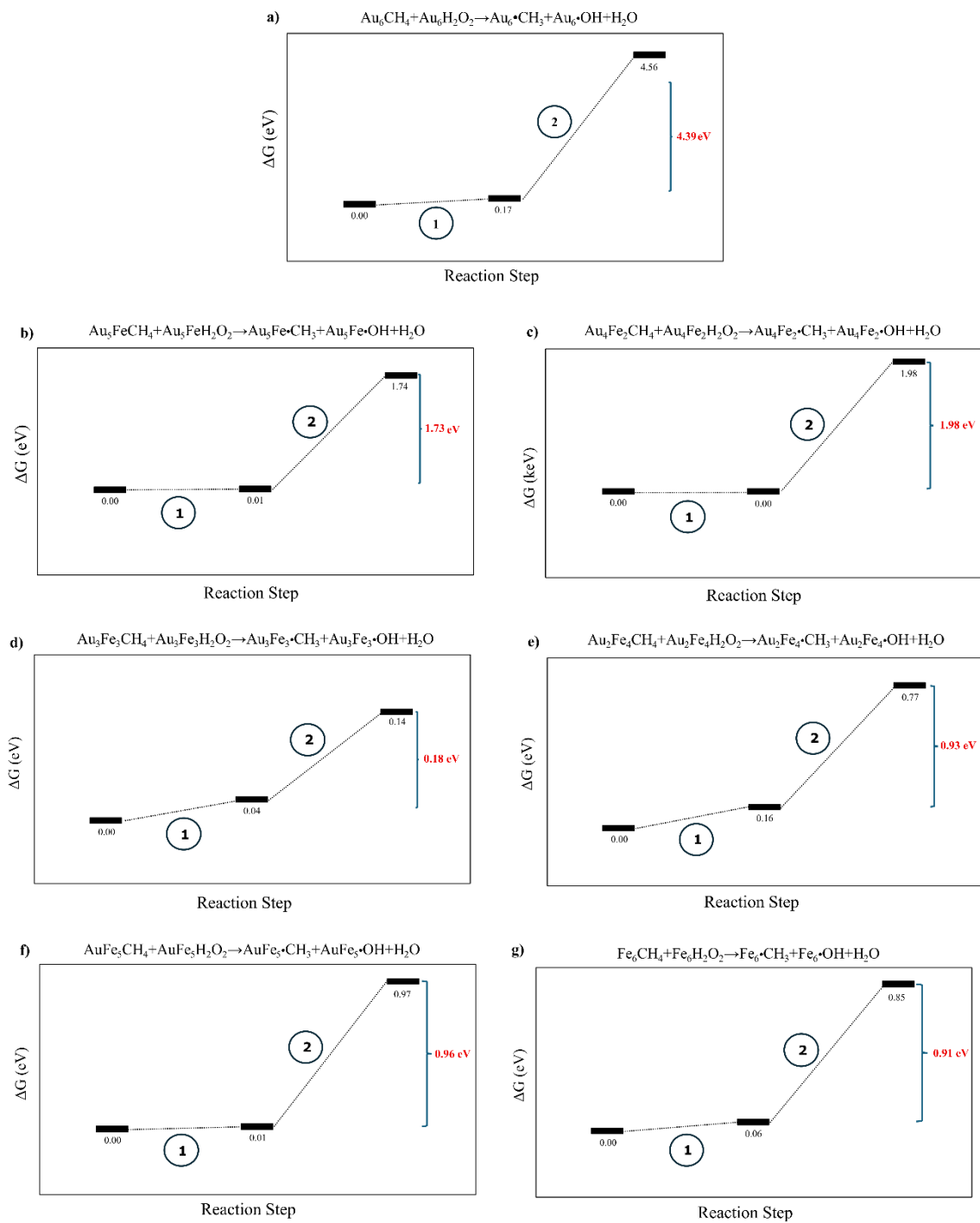


Figure S5. Barrier energies for the obtained radicals.

2.2. Vibrational modes

Geometries of the adsorbed radical species were taken directly from their closed-shell molecular counterparts, with no significant structural changes observed. Distances and vibrational frequencies for $\text{Au}_x\text{Fe}_y\text{-}\bullet\text{CH}_3$ and $\text{Au}_x\text{Fe}_y\text{-}\bullet\text{OH}$ systems ($x + y = 6$) correlate well (Tables S7 and S10), indicating stronger or weaker interactions depending on the adsorption site. For instance, the $\text{HO}\bullet\text{-FeAu}_5$ distance (1.794 Å, $\nu_{\text{Fe-O}} = 685 \text{ cm}^{-1}$) is slightly shorter than that observed for $\bullet\text{CH}_3\text{-Au}_5\text{Fe}$ (1.986 Å, $\nu_{\text{Au-C}} = 497 \text{ cm}^{-1}$). As the proportion of iron atoms increases, both radicals preferentially interact with the Fe centers, forming shorter and therefore stronger bonds in $\text{HO}\bullet\text{-clusters}$, ranging from 1.794 Å in $\text{HO}\bullet\text{-FeAu}_5$ to 1.956 Å in $\text{Au}_3\text{Fe}_3\text{HO}\bullet$, the latter is associated with $\nu_{\text{Fe-O}} = 529 \text{ cm}^{-1}$. In contrast, the shortest distance in $\bullet\text{CH}_3\text{-clusters}$ is 1.996 Å for $\text{Au}_4\text{Fe}_2\text{-}\bullet\text{CH}_3$ ($\nu_{\text{Fe-C}} = 508 \text{ cm}^{-1}$), while the largest is 2.165 Å in $\text{Au}_3\text{Fe}_3\text{-}\bullet\text{CH}_3$, where the radical binds with two iron atoms having a lower stretching frequency of 443 cm^{-1} . Unlike the methyl radical, which induces only minor structural rearrangements due to its limited polarizability and reactivity, the hydroxyl radical, with a strongly negative oxygen center, forms considerably stronger interactions with the positively charged metal sites.

Table S10. Vibrational modes of main bonds.

Species	$\bar{\nu}_{\text{maxAu-Au}}$	$\bar{\nu}_{\text{maxFe-Fe}}$	$\bar{\nu}_{\text{maxAu-Fe}}$	$\bar{\nu}_{\text{maxAu-C}}$	$\bar{\nu}_{\text{maxAu-O}}$	$\bar{\nu}_{\text{maxFe-C}}$	$\bar{\nu}_{\text{maxFe-O}}$	$\bar{\nu}_{\text{maxC-O}}$	$\bar{\nu}_{\text{maxO-O}}$	$\bar{\nu}_{\text{maxC-H}}$	$\bar{\nu}_{\text{maxO-H}}$
CH ₄										3168	
H ₂ O ₂									1007		3807
H ₃ C•										3311	
HO•											3749
H•											
CH ₃ OH								1108		3132	3880
H ₂ O											3963
Bare clusters											
Au ₆	178										
Au ₅ Fe	131		269								
Au ₄ Fe ₂			288								
Au ₃ Fe ₃		265	217								
Au ₂ Fe ₄		272	237								
AuFe ₅		266	205								
Fe ₆		273									
CH ₄ adsorption											
Au ₆ CH ₄	179			62						3162	
Au ₅ FeCH ₄	131		263	-----						3177	
Au ₄ Fe ₂ CH ₄			272			59				3175	
Au ₃ Fe ₃ CH ₄			269			145				3176	
Au ₂ Fe ₄ CH ₄		222	236			149				3162	
AuFe ₅ CH ₄		267	133			91				3148	
Fe ₆ CH ₄		269				41				3153	
H ₂ O ₂ adsorption											
Au ₆ H ₂ O ₂	176				186				998		3797
Au ₅ FeH ₂ O ₂	134		246			288			976		3750
Au ₄ Fe ₂ H ₂ O ₂			251			287			972		3722
Au ₃ Fe ₃ H ₂ O ₂			251			298			983		3751
Au ₂ Fe ₄ H ₂ O ₂		251	232			280			973		3764

AuFe ₅ H ₂ O ₂		205	191				295		970		3773
Fe ₆ H ₂ O ₂		213					278		980		3773
Methyl radical Formation											
Au ₆ H ₃ C•	176			484						3187	
Au ₅ FeH ₃ C•	153		267	527						3164	
Au ₄ Fe ₂ H ₃ C•	98	221	242			508				3151	
Au ₃ Fe ₃ H ₃ C•		208	249			443				3104	
Au ₂ Fe ₄ H ₃ C•		266	266			514				3101	
AuFe ₅ H ₃ C•		286	234			446				3101	
Fe ₆ H ₃ C•		261				504				3088	
Hydroxyl radical Formation											
Au ₆ HO•	178				497						3836
Au ₅ FeHO•	115		152			685					3928
Au ₄ Fe ₂ HO•		220	242			676					3923
Au ₃ Fe ₃ HO•		145	198			529					3908
Au ₂ Fe ₄ HO•		265	244			654					3920
AuFe ₅ HO•		247	115			656					3937
Fe ₆ HO•		264				651					3934
Transition States											
(Au ₆ OH–CH ₃) _{ts}	173			401	447			-418		3291	3822
(Au ₅ FeOH–CH ₃) _{ts}	120		174			634		-525		3315	3873
(Au ₄ Fe ₂ OH–CH ₃) _{ts}			243			468		-561		3318	3825
(Au ₃ Fe ₃ OH–CH ₃) _{ts}		181	263			346		-741		3294	3658
(Au ₂ Fe ₄ OH–CH ₃) _{ts}		264	241			423		-599		3313	3815
(AuFe ₅ OH–CH ₃) _{ts}		120	222			449		-717		3303	3900
(Fe ₆ OH–CH ₃) _{ts}		278				447		-401		3325	3844
Cluster – Methanol											
Au ₆ –CH ₃ OH	175				213			1058		3166	3860
Au ₅ Fe–CH ₃ OH	133		240			294		1057		3168	3822
Au ₄ Fe ₂ –CH ₃ OH			284			330		1045		3173	3797
Au ₃ Fe ₃ –CH ₃ OH		209	262			307		1046		3173	3844
Au ₂ Fe ₄ –CH ₃ OH		257	229			289		1051		3165	3743
AuFe ₅ –CH ₃ OH		227	179			306		1049		3172	3847
Fe ₆ –CH ₃ OH		212				292		1049		3171	3832

2.3. Charges

Upon formation of the radical complexes, highest electronic transferences were found for the hydroxyl radicals, with the oxygen site being highly susceptible to electrophilic attack and carrying a total charge between –0.892 and –1.087 e (Table S11), similar to the oxygen in a water molecule (–0.913 e). The methyl radical (•CH₃) shows a negatively charged carbon atom (–0.500 e), indicating that the unpaired electron is primarily localized at that site.

Table S11. Total and mean charges by atom in molecules, clusters and complexes.

Species	Q _{Au} (e)	Q _{Fe} (e)	$\bar{q}_{Au}$ (e)	$\bar{q}_{Fe}$ (e)	q _C (e)	Q _O (e)	$\bar{q}_O$ (e)	Q _H (e)	$\bar{q}_H$ (e)
CH ₄					-0.857			0.857	0.214
H ₂ O ₂						-0.893	-0.446	0.893	0.446

H ₃ C•					-0.500			0.500	0.1665
HO•						-0.397		0.397	
H•								0.000	0.000
CH ₃ OH					-0.266	-0.686		0.951	0.238
H ₂ O						-0.913		0.914	0.457
Bare clusters									
Au ₆	0.000		0.000						
Au ₅ Fe	-0.301	0.301	-0.060						
Au ₄ Fe ₂	-0.820	0.820	-0.205	0.410					
Au ₃ Fe ₃	-0.703	0.703	-0.234	0.234					
Au ₂ Fe ₄	-0.502	0.502	-0.251	0.126					
AuFe ₅	-0.457	0.457		0.091					
Fe ₆		0.000		0.000					
CH ₄ adsorption									
Au ₆ CH ₄	-0.015		-0.005		-0.881			0.896	0.220
Au ₅ FeCH ₄	-0.286	0.263	-0.057		-0.854			0.879	0.220
Au ₄ Fe ₂ CH ₄	-0.738	0.708	-0.185	0.354	-0.860			0.890	0.223
Au ₃ Fe ₃ CH ₄	-0.652	0.589	-0.217	0.196	-0.887			0.940	0.235
Au ₂ Fe ₄ CH ₄	-0.501	0.435	-0.251	0.109	-0.904			0.970	0.243
AuFe ₅ CH ₄	-0.468	0.335		0.067	-0.908			0.953	0.238
Fe ₆ CH ₄		-0.016		-0.003	-0.889			0.904	0.226
H ₂ O ₂ adsorption									
Au ₆ H ₂ O ₂	-0.055		-0.009			-0.882	-0.441	0.935	0.468
Au ₅ FeH ₂ O ₂	-0.311	0.201	-0.062			-0.854	-0.427	0.963	0.482
Au ₄ Fe ₂ H ₂ O ₂	-0.833	0.729	-0.208	0.365		-0.861	-0.431	0.964	0.482
Au ₃ Fe ₃ H ₂ O ₂	-0.643	0.557	-0.214	0.186		-0.868	-0.434	0.953	0.477
Au ₂ Fe ₄ H ₂ O ₂	-0.560	0.480	-0.280	0.120		-0.867	-0.434	0.948	0.474
AuFe ₅ H ₂ O ₂	-0.449	0.338		0.090		-0.888	-0.444	0.949	0.475
Fe ₆ H ₂ O ₂		-0.058		-0.010		-0.887	-0.444	0.944	0.472
Methyl radical Formation									
Au ₆ H ₃ C•	0.185		0.031		-0.832			0.647	0.216
Au ₅ FeH ₃ C•	-0.132	0.689	-0.027		-0.968			0.666	0.222
Au ₄ Fe ₂ H ₃ C•	-0.369	0.699	-0.092	0.345	-1.008			0.674	0.225
Au ₃ Fe ₃ H ₃ C•	-0.575	1.029	-0.192	0.343	-1.169			0.715	0.238
Au ₂ Fe ₄ H ₃ C•	-0.453	0.964	-0.227	0.241	-1.169			0.658	0.219
AuFe ₅ H ₃ C•	-0.211	0.621		0.124	-1.119			0.707	0.236
Fe ₆ H ₃ C•		0.524		0.087	-1.171			0.648	0.216
Hydroxyl radical Formation									
Au ₆ HO•	0.420		0.070			-0.873		0.453	
Au ₅ FeHO•	0.066	0.467	0.132			-1.007		0.475	
Au ₄ Fe ₂ HO•	-0.326	0.881	-0.082	0.441		-1.028		0.473	
Au ₃ Fe ₃ HO•	-0.649	1.243	-0.216	0.414		-1.089		0.494	
Au ₂ Fe ₄ HO•	-0.360	0.909	-0.180	0.227		-1.016		0.467	
AuFe ₅ HO•	-0.467	1.059		0.212		-1.059		0.467	
Fe ₆ HO•		0.588		0.098		-1.056		0.466	
Transition States									
(Au ₆ OH–CH ₃) _{ts}	0.165		0.028		-0.866	-0.412		1.113	0.278
(Au ₅ FeOH–CH ₃) _{ts}	-0.167	0.483	-0.033		-0.439	-0.927		1.050	0.263
(Au ₄ Fe ₂ OH–CH ₃) _{ts}	-0.535	0.858	-0.134	0.429	-0.389	-0.969		1.036	0.259
(Au ₃ Fe ₃ OH–CH ₃) _{ts}	-0.694	0.985	-0.231	0.328	-0.413	-0.961		1.082	0.271
(Au ₂ Fe ₄ OH–CH ₃) _{ts}	-0.438	0.771	-0.219	0.193	-0.410	-0.949		1.026	0.256
(AuFe ₅ OH–CH ₃) _{ts}	-0.509	0.855		0.171	-0.431	-0.940		1.025	0.257

(Fe ₆ OH-CH ₃) _{ts}		0.372		0.062	-0.411	-0.967		1.005	0.251
Cluster – Methanol									
Au ₆ -CH ₃ OH	-0.066		-0.011		-0.265	-0.711		1.042	0.261
Au ₅ Fe-CH ₃ OH	-0.456	0.322	-0.091		-0.263	-0.692		1.089	0.272
Au ₄ Fe ₂ -CH ₃ OH	-0.957	0.828	-0.239	0.414	-0.268	-0.706		1.104	0.276
Au ₃ Fe ₃ -CH ₃ OH	-0.815	0.702	-0.272	0.234	-0.267	-0.706		1.085	0.271
Au ₂ Fe ₄ -CH ₃ OH	-0.590	0.491	-0.295	0.123	-0.270	-0.704		1.073	0.268
AuFe ₅ -CH ₃ OH	-0.502	0.394		0.079	-0.269	-0.703		1.079	0.270
Fe ₆ -CH ₃ OH		-0.097		-0.016	-0.269	-0.705		1.072	0.268

3. Transition States and Products

Transition states exhibit total oxygen charges ranging from -0.927 to -0.969 e for Au₅Fe and Au₄Fe₂, respectively. Only the Au₆ TS displays a lower oxygen charge (-0.412 e), where the unpaired electron is mainly on the carbon atom (-0.866 e), with •OH interacting with the six-gold-atom cluster. For the other species, carbon charges range from -0.389 to -0.429 e, being most negative in Au₅Fe and least in Au₄Fe₂ (Figure S6). In the product complexes, charge transfer occurs predominantly to the gold atoms of the cluster, with values shifting from the TS range of -0.694 to 0.165 e to -0.957 to -0.066 e in the products. Carbon atoms retain slightly negative charges (-0.271 to -0.263 e), while oxygen atoms maintain significant negative charges (-0.713 to -0.690 e). Iron atoms display similar charge ranges in both TSs and products (-0.098 to 0.958 e), retaining residual charge, especially in the pure iron clusters.

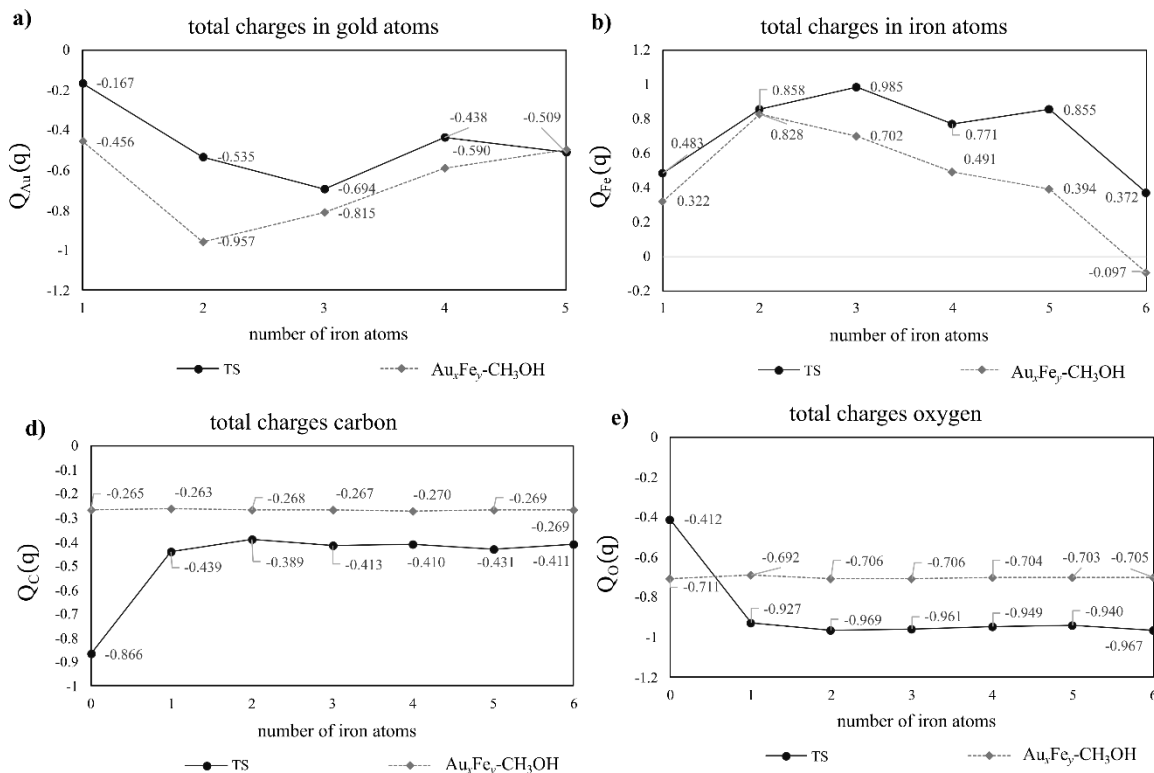


Figure S6. Total charges of a) gold atoms, b) iron atoms, c) carbon atom, and d) oxygen atom in transition states (TSs), and products

The unpaired electron in $Au_xFe_y\cdot CH_3$ and $Au_xFe_y\cdot OH$ is mainly localized on the carbon or oxygen atoms, respectively, which favors the oxygen site as the preferred one for the methyl adsorption. Contrarily, the cluster-methyl interactions leads to a more delocalized charge distribution over the entire system (Table S11). Even more, the magnetic analysis indicates that the average moment on the gold atoms ($\bar{\mu}_{Au}$) have an increase, reaching $0.287 \mu_B$ in $AuFe_5\cdot OH$, which are coupled ferromagnetically with the higher moments on the iron sites ($\bar{\mu}_{Fe} = 3.517 \mu_B$) and with that of the oxygen atom ($\bar{\mu}_O = 0.122 \mu_B$), producing a total magnetic moment of $18.00 \mu_B$. Which should be compared with those of bare $AuFe_5$ and $AuFe_5-H_2O_2$, both possessing $17.00 \mu_B$. Such spin distribution highlights the ferromagnetic behavior of the whole $AuFe_5\cdot OH$ complex. Regarding radical formation, the lowest energy barrier was observed for the Au_3Fe_3 composition (Figure S5), where the obtained $Au_3Fe_3\cdot OH + \cdot CH_3$ transition state has the most exergonic value, and resultant spontaneity, as compared with its $Au_3Fe_3\cdot CH_3 + \cdot OH$ counterpart (Figures S7 and S8).

For spin distribution, the carbon atom of the free methyl radical carries the largest spin ($1.134 \mu_B$, Table S8), while its hydrogens show negative spin ($-0.045 \mu_B$), confirming the unpaired electron

localization. Hydroxyl radical complexes exhibit substantial Mulliken spin on the cluster (3.198–3.682 μ_B), with the oxygen atoms carrying 0.087–0.323 μ_B , enabling them to accept the unpaired electron from the methyl radical. This is reflected in the transition states and products (Table S8), where Mulliken spin on oxygen and carbon is lower than in isolated species (–0.063 to 0.096 μ_B for oxygen in TSs and products, –0.719 to 0.841 μ_B for carbon in TSs), stabilizing in the products (–0.006 to 0.002 μ_B). These results indicate effective spin transfer from the methyl radical to the hydroxyl radical complexes, facilitating methanol formation and stabilization.

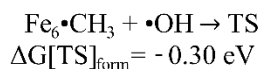
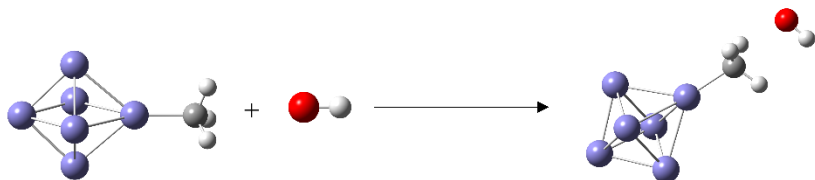
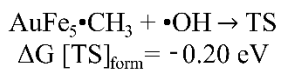
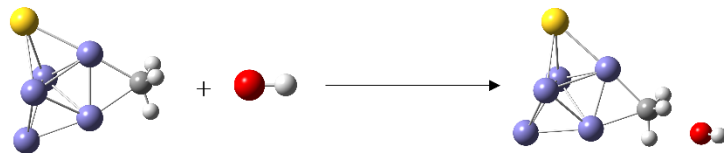
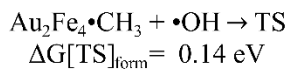
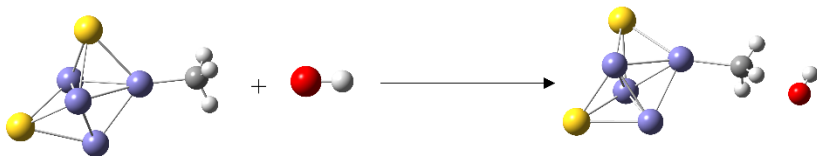
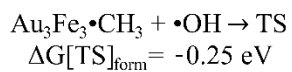
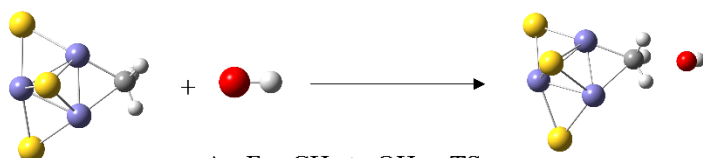
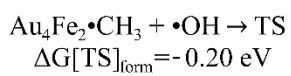
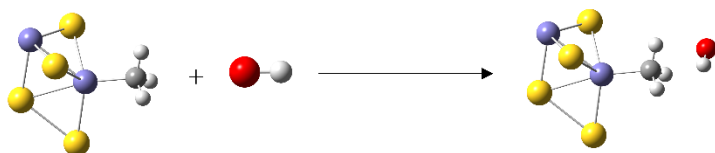
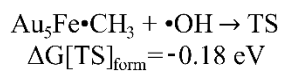
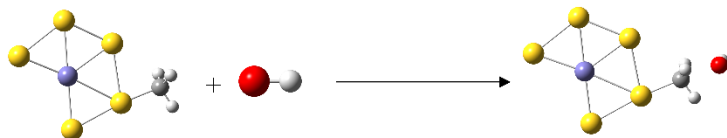
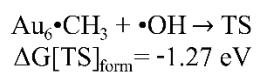
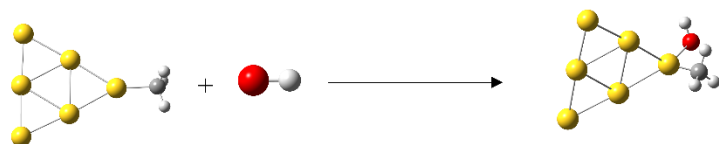


Figure S7. $(\text{Au}_x\text{Fe}_y\text{CH}_3\text{--OH})_{\text{ts}}$ formation free energies, where $x + y = 6$.

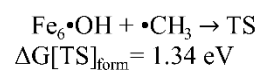
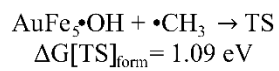
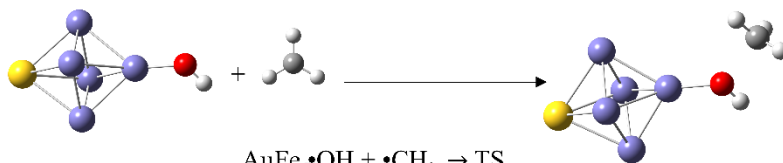
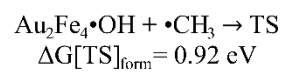
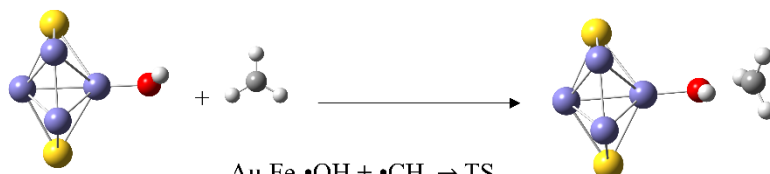
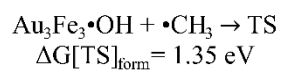
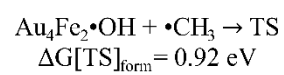
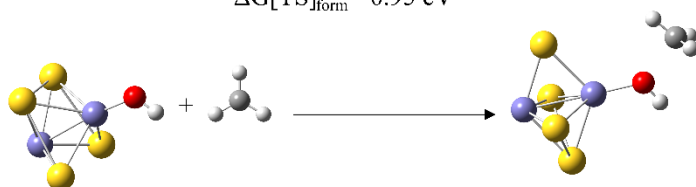
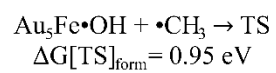
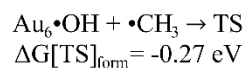
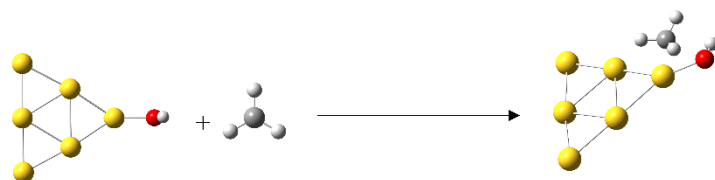


Figure S8. $(\text{Au}_x\text{Fe}_y\text{OH}-\text{CH}_3)_{\text{ts}}$ formation free energies, where $x + y = 6$.

Initial structures, in steps 1 and 2 (Figure 2) were discussed previously. In the absence of a cluster, the free energy required to generate both radical species is 5.98 eV, a barrier that is significantly reduced in the presence of Au_xFe_y clusters, ranging from 4.56 eV for pure gold clusters down to 0.23 eV for the 50:50 Au:Fe composition. Once the cluster–radical complexes are formed, the second radical is added, leading to different transition states for $\text{Au}_x\text{Fe}_y\text{CH}_3-\text{OH}$ and $\text{Au}_x\text{Fe}_y\text{OH}-\text{CH}_3$ (Figures S7 and S8, respectively). The imaginary frequencies computed for $\text{Au}_x\text{Fe}_y\text{CH}_3-\text{OH}$ are not aligned with the reaction coordinate for methanol formation. In contrast, all $\text{Au}_x\text{Fe}_y\text{OH}-\text{CH}_3$ transition states exhibit imaginary frequencies along the reaction coordinate (ranging from -741 to -401 cm^{-1}) to reach methanol synthesis (Table S10), consistent with the products obtention (Figure S9).

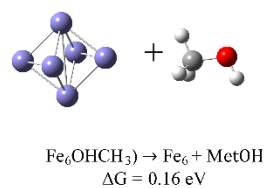
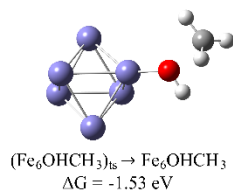
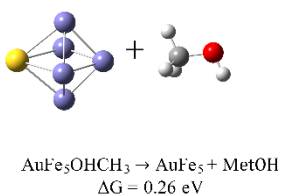
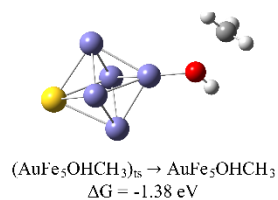
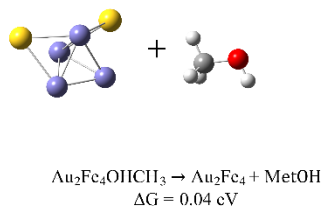
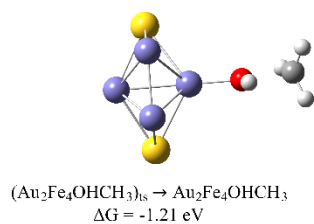
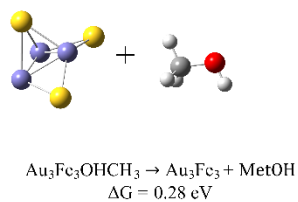
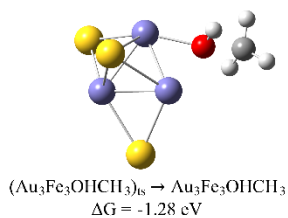
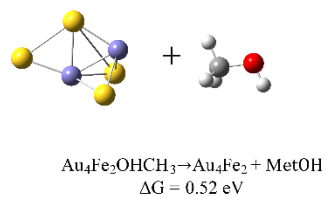
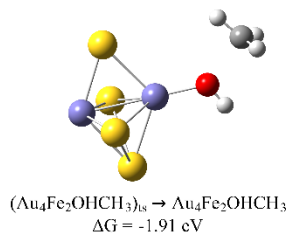
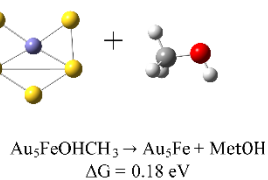
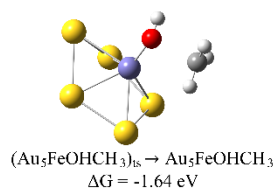
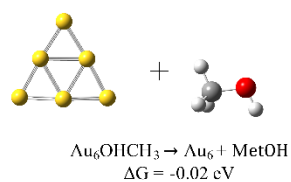
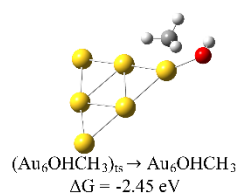


Figure S9. $(\text{Au}_x\text{Fe}_y\text{OH}-\text{CH}_3)_{\text{ts}} \rightarrow \text{Au}_x\text{Fe}_y-\text{CH}_3\text{OH}$ and $\text{Au}_x\text{Fe}_y-\text{OHCH}_3 \rightarrow (\text{Au}_x\text{Fe}_y + \text{CH}_3\text{OH})$ dissociation free Energies, where $x + y = 6$.

Metal–metal distances in $\text{Au}_x\text{Fe}_y\cdot\text{CH}_3+\cdot\text{OH}$, and $\text{Au}_x\text{Fe}_y\cdot\text{OH}+\cdot\text{CH}_3$ transition states (TSs) fall within similar ranges as those observed in the previous interactions (2.470–2.980 Å). Even in the subsequent steps, leading to the relaxed products $\text{Au}_x\text{Fe}_y-\text{OHCH}_3$, these distances remain within the same range (Table S7). The distances between the clusters and the oxygen atom in the TSs vary from 1.851 to 2.111 Å, increasing upon formation of the cluster–methanol complexes (2.097–2.411 Å), which indicates weakening of the metal–oxygen bonds (Figure S10a). Notably, the carbon–oxygen interaction in the TSs ranges from 1.891 to 2.115 Å. In the products $\text{Au}_x\text{Fe}_y-\text{OHCH}_3$, the C–O bond lengths are within 1.425–1.433 Å (Figure S10b), consistent with typical methanol bond distances [Table S1].

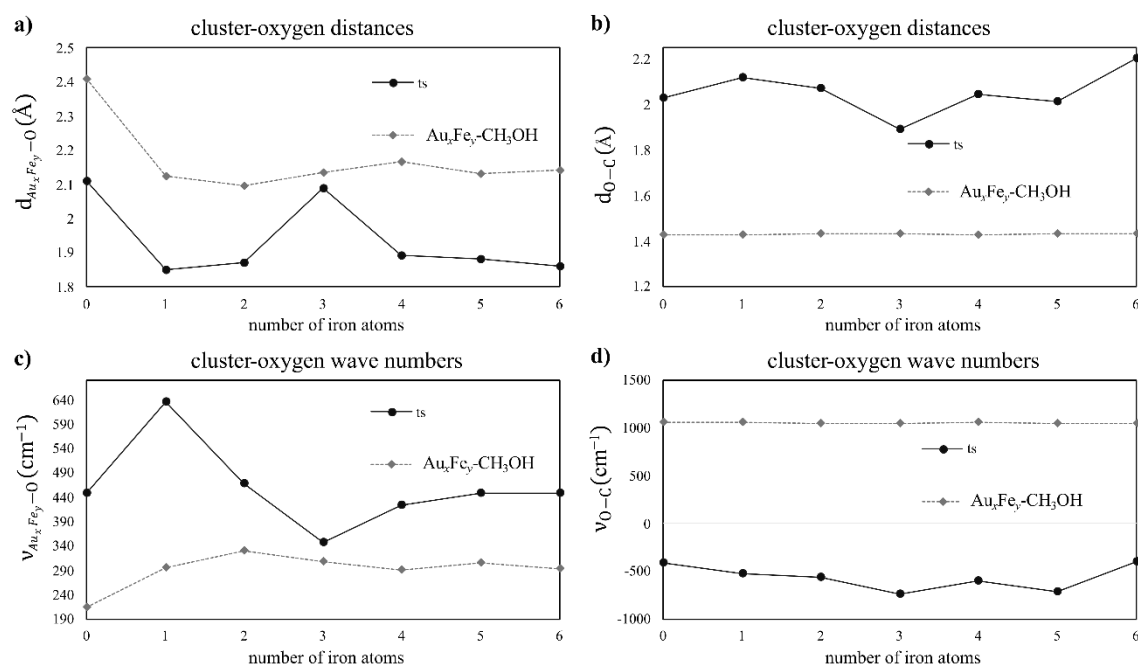


Figure S10. a, c) $\text{Au}_x\text{Fe}_y-\text{O}$, b, d) O–C distances and frequencies trends in transition states and products as the proportion of iron atoms increases.

The weakening of the $\text{Au}_x\text{Fe}_y-\text{O}$ bonds is also reflected in their vibrational frequencies, decreasing from 447–634 cm^{-1} in the transition states to 213–330 cm^{-1} in the product complexes (Figures S10c and S10d). As expected, the C–O bonds are strongest in the product states, showing typical methanol stretching frequencies of 1045–1060 cm^{-1} [51].

4. Methanol Formation Reaction Path

As mentioned, the most energetically favorable reaction pathway proceeds through the adsorption and dissociation of the hydrogen peroxide molecule on the cluster. Once the $\text{Au}_x\text{Fe}_y\text{HO}\cdot$ radical is formed, a free $\cdot\text{CH}_3$ species reacts with the complex, reaching a transition state that leads to $\text{Au}_x\text{Fe}_y\text{OHCH}_3$ formation. Subsequently, a small energy input is required for the cleavage of the $\text{Au}_x\text{Fe}_y\text{OHCH}_3$ bond. Transition states relaxed product geometries, and bond cleavage energies are summarized in Figures S7–S9, and Table S12. Figure S11 illustrates the six-step reaction pathway for the $\text{Au}_3\text{Fe}_3\text{CH}_4 + \text{Au}_3\text{Fe}_3\text{H}_2\text{O}_2 \rightarrow 2\text{Au}_3\text{Fe}_3 + \text{CH}_3\text{OH} + \text{H}_2\text{O}$ process.

Table S12. Free energies.

Organic Species	ΔE (eV)	ΔG (eV)
$\text{CH}_4 \rightarrow \text{H}_3\text{C}\cdot + \text{H}\cdot$ ($\Delta E_{\text{H}_3\text{C}\cdot\text{-formation}}$)	4.33	4.29
$\text{H}_2\text{O}_2 \rightarrow \text{HO}\cdot + \text{HO}\cdot$ ($\Delta E_{\text{HO}\cdot\text{-formation}}$)	2.01	1.69
$\text{H}_3\text{C}\cdot + \text{H}\cdot \rightarrow \text{CH}_3\text{OH}$ ($\Delta E_{\text{CH}_3\text{OH-formation}}$)	-3.80	-3.45
$\text{HO}\cdot + \text{HO}\cdot \rightarrow \text{H}_2\text{O}$ ($\Delta E_{\text{H}_2\text{O-formation}}$)	-4.84	-4.86
$\text{Au}_6 + \text{CH}_4 \rightarrow \text{Au}_6\text{CH}_4$ ($\Delta E_{\text{Au}_6\text{CH}_4\text{-adsorption}}$)	-0.06	0.07
$\text{Au}_5\text{Fe} + \text{CH}_4 \rightarrow \text{Au}_5\text{FeCH}_4$ ($\Delta E_{\text{Au}_5\text{FeCH}_4\text{-adsorption}}$)	-0.05	0.13
$\text{Au}_4\text{Fe}_2 + \text{CH}_4 \rightarrow \text{Au}_4\text{Fe}_2\text{CH}_4$ ($\Delta E_{\text{Au}_4\text{Fe}_2\text{CH}_4\text{-adsorption}}$)	-0.11	0.15
$\text{Au}_3\text{Fe}_3 + \text{CH}_4 \rightarrow \text{Au}_3\text{Fe}_3\text{CH}_4$ ($\Delta E_{\text{Au}_3\text{Fe}_3\text{CH}_4\text{-adsorption}}$)	-0.08	0.18
$\text{Au}_2\text{Fe}_4 + \text{CH}_4 \rightarrow \text{Au}_2\text{Fe}_4\text{CH}_4$ ($\Delta E_{\text{Au}_2\text{Fe}_4\text{CH}_4\text{-adsorption}}$)	-0.11	0.15
$\text{AuFe}_5 + \text{CH}_4 \rightarrow \text{AuFe}_5\text{CH}_4$ ($\Delta E_{\text{AuFe}_5\text{CH}_4\text{-adsorption}}$)	-0.07	0.15
$\text{Fe}_6 + \text{CH}_4 \rightarrow \text{Fe}_6\text{CH}_4$ ($\Delta E_{\text{Fe}_6\text{CH}_4\text{-adsorption}}$)	-0.03	0.13
$\text{Au}_6 + \text{H}_2\text{O}_2 \rightarrow \text{Au}_6\text{H}_2\text{O}_2$ ($\Delta E_{\text{Au}_6\text{H}_2\text{O}_2\text{-adsorption}}$)	-0.24	0.10
$\text{Au}_5\text{Fe} + \text{H}_2\text{O}_2 \rightarrow \text{Au}_5\text{FeH}_2\text{O}_2$ ($\Delta E_{\text{Au}_5\text{FeH}_2\text{O}_2\text{-adsorption}}$)	-0.59	-0.12
$\text{Au}_4\text{Fe}_2 + \text{H}_2\text{O}_2 \rightarrow \text{Au}_4\text{Fe}_2\text{H}_2\text{O}_2$ ($\Delta E_{\text{Au}_4\text{Fe}_2\text{H}_2\text{O}_2\text{-adsorption}}$)	-0.62	-0.16

$\text{Au}_3\text{Fe}_3 + \text{H}_2\text{O}_2 \rightarrow \text{Au}_3\text{Fe}_3\text{H}_2\text{O}_2$ ($\Delta E_{\text{Au}_3\text{Fe}_3\text{H}_2\text{O}_2-\text{adsorption}}$)	-0.57	-0.14
$\text{Au}_2\text{Fe}_4 + \text{H}_2\text{O}_2 \rightarrow \text{Au}_2\text{Fe}_4\text{H}_2\text{O}_2$ ($\Delta E_{\text{Au}_2\text{Fe}_4\text{H}_2\text{O}_2-\text{adsorption}}$)	-0.43	0.01
$\text{AuFe}_5 + \text{H}_2\text{O}_2 \rightarrow \text{AuFe}_5\text{H}_2\text{O}_2$ ($\Delta E_{\text{AuFe}_5\text{H}_2\text{O}_2-\text{adsorption}}$)	-0.58	-0.14
$\text{Fe}_6 + \text{H}_2\text{O}_2 \rightarrow \text{Fe}_6\text{H}_2\text{O}_2$ ($\Delta E_{\text{Fe}_6\text{H}_2\text{O}_2-\text{adsorption}}$)	-0.51	-0.07
$\text{Au}_6\text{CH}_4 \rightarrow \text{Au}_6\text{H}_3\text{C}\cdot + \text{H}\cdot$ ($\Delta E_{\text{Au}_6\text{CH}_3\cdot-\text{formation}}$)	3.41	3.50
$\text{Au}_5\text{FeCH}_4 \rightarrow \text{Au}_5\text{FeH}_3\text{C}\cdot + \text{H}\cdot$ ($\Delta E_{\text{Au}_5\text{CH}_3\cdot-\text{formation}}$)	2.68	2.85
$\text{Au}_4\text{Fe}_2\text{CH}_4 \rightarrow \text{Au}_4\text{Fe}_2\text{H}_3\text{C}\cdot + \text{H}\cdot$ ($\Delta E_{\text{Au}_4\text{Fe}_2\text{CH}_3\cdot-\text{formation}}$)	3.00	3.10
$\text{Au}_3\text{Fe}_3\text{CH}_4 \rightarrow \text{Au}_3\text{Fe}_3\text{H}_3\text{C}\cdot + \text{H}\cdot$ ($\Delta E_{\text{Au}_3\text{Fe}_3\text{CH}_3\cdot-\text{formation}}$)	2.01	2.14
$\text{Au}_2\text{Fe}_4\text{CH}_4 \rightarrow \text{Au}_2\text{Fe}_4\text{H}_3\text{C}\cdot + \text{H}\cdot$ ($\Delta E_{\text{Au}_2\text{Fe}_4\text{CH}_3\cdot-\text{formation}}$)	2.38	2.44
$\text{AuFe}_5\text{CH}_4 \rightarrow \text{AuFe}_5\text{H}_3\text{C}\cdot + \text{H}\cdot$ ($\Delta E_{\text{AuFe}_5\text{CH}_3\cdot-\text{formation}}$)	2.37	2.55
$\text{Fe}_6\text{CH}_4 \rightarrow \text{Fe}_6\text{H}_3\text{C}\cdot + \text{H}\cdot$ ($\Delta E_{\text{Fe}_6\text{CH}_3\cdot-\text{formation}}$)	2.43	2.55
$\text{Au}_6\text{H}_2\text{O}_2 \rightarrow \text{Au}_6\text{HO}\cdot + \text{HO}\cdot$ ($\Delta E_{\text{Au}_6\text{HO}\cdot-\text{formation}}$)	1.22	0.89
$\text{Au}_5\text{FeH}_2\text{O}_2 \rightarrow \text{Au}_5\text{FeHO}\cdot + \text{HO}\cdot$ ($\Delta E_{\text{Au}_5\text{FeHO}\cdot-\text{formation}}$)	-0.68	-1.12
$\text{Au}_4\text{Fe}_2\text{H}_2\text{O}_2 \rightarrow \text{Au}_4\text{Fe}_2\text{HO}\cdot + \text{HO}\cdot$ ($\Delta E_{\text{Au}_4\text{Fe}_2\text{HO}\cdot-\text{formation}}$)	-0.72	-1.13
$\text{Au}_3\text{Fe}_3\text{H}_2\text{O}_2 \rightarrow \text{Au}_3\text{Fe}_3\text{HO}\cdot + \text{HO}\cdot$ ($\Delta E_{\text{Au}_3\text{Fe}_3\text{HO}\cdot-\text{formation}}$)	-1.61	-1.95
$\text{Au}_2\text{Fe}_4\text{H}_2\text{O}_2 \rightarrow \text{Au}_2\text{Fe}_4\text{HO}\cdot + \text{HO}\cdot$ ($\Delta E_{\text{Au}_2\text{Fe}_4\text{HO}\cdot-\text{formation}}$)	-1.13	-1.51
$\text{AuFe}_5\text{H}_2\text{O}_2 \rightarrow \text{AuFe}_5\text{HO}\cdot + \text{HO}\cdot$ ($\Delta E_{\text{AuFe}_5\text{HO}\cdot-\text{formation}}$)	-1.15	-1.59
$\text{Fe}_6\text{H}_2\text{O}_2 \rightarrow \text{Fe}_6\text{HO}\cdot + \text{HO}\cdot$ ($\Delta E_{\text{Fe}_6\text{HO}\cdot-\text{formation}}$)	-1.22	-1.65
$(\text{Au}_6\text{OH} - \text{CH}_3)_{\text{ts}}$ ($\Delta E_{\text{barrier energy}}$)	-0.74	-0.27
$(\text{Au}_5\text{FeOH} - \text{CH}_3)_{\text{ts}}$ ($\Delta E_{\text{barrier energy}}$)	0.54	0.95
$(\text{Au}_4\text{Fe}_2\text{OH} - \text{CH}_3)_{\text{ts}}$ ($\Delta E_{\text{barrier energy}}$)	0.60	0.92

$(\text{Au}_3\text{Fe}_3\text{OH} - \text{CH}_3)_{\text{ts}}$ ($\Delta E_{\text{barrier energy}}$)	0.96	1.35
$(\text{Au}_2\text{Fe}_4\text{OH} - \text{CH}_3)_{\text{ts}}$ ($\Delta E_{\text{barrier energy}}$)	0.57	0.92
$(\text{AuFe}_5\text{OH} - \text{CH}_3)_{\text{ts}}$ ($\Delta E_{\text{barrier energy}}$)	0.70	1.09
$(\text{Fe}_6\text{OH} - \text{CH}_3)_{\text{ts}}$ ($\Delta E_{\text{barrier energy}}$)	0.99	1.34
$(\text{Au}_6\text{OH} - \text{CH}_3)_{\text{ts}} \rightarrow \text{Au}_6 - \text{OHCH}_3$ ($\Delta E_{\text{Au}_6\text{OHCH}_3\text{-formation}}$)	-2.38	-2.45
$(\text{Au}_5\text{FeOH} - \text{CH}_3)_{\text{ts}} \rightarrow \text{Au}_5\text{Fe} - \text{OHCH}_3$ ($\Delta E_{\text{Au}_5\text{FeOHCH}_3\text{-formation}}$)	-1.68	-1.64
$(\text{Au}_4\text{Fe}_2\text{OH} - \text{CH}_3)_{\text{ts}} \rightarrow \text{Au}_4\text{Fe}_2 - \text{OHCH}_3$ ($\Delta E_{\text{Au}_4\text{Fe}_2\text{OHCH}_3\text{-formation}}$)	-1.95	-1.91
$(\text{Au}_3\text{Fe}_3\text{OH})_{\text{ts}} \rightarrow \text{Au}_3\text{Fe}_3 - \text{OHCH}_3$ ($\Delta E_{\text{Au}_3\text{Fe}_3\text{OHCH}_3\text{-formation}}$)	-1.21	-1.28
$(\text{Au}_2\text{Fe}_4\text{OH})_{\text{ts}} \rightarrow \text{Au}_2\text{Fe}_4 - \text{OHCH}_3$ ($\Delta E_{\text{Au}_2\text{Fe}_4\text{OHCH}_3\text{-formation}}$)	-1.24	-1.21
$(\text{AuFe}_5\text{OH} - \text{CH}_3)_{\text{ts}} \rightarrow \text{AuFe}_5 - \text{OHCH}_3$ ($\Delta E_{\text{AuFe}_5\text{OHCH}_3\text{-formation}}$)	-1.38	-1.38
$(\text{Fe}_6\text{OH} - \text{CH}_3)_{\text{ts}} \rightarrow \text{Fe}_6 - \text{OHCH}_3$ ($\Delta E_{\text{Fe}_6\text{OHCH}_3\text{-formation}}$)	-1.57	-1.52
$\text{Au}_6 - \text{OHCH}_3 \rightarrow \text{Au}_6 + \text{OHCH}_3$ ($\Delta E_{\text{dissociation}}$)	0.36	-0.02
$\text{Au}_5\text{Fe} - \text{OHCH}_3 \rightarrow \text{Au}_5\text{Fe} + \text{OHCH}_3$ ($\Delta E_{\text{dissociation}}$)	0.63	0.18
$\text{Au}_4\text{Fe}_2 - \text{OHCH}_3 \rightarrow \text{Au}_4\text{Fe}_2 + \text{OHCH}_3$ ($\Delta E_{\text{dissociation}}$)	0.91	0.52
$\text{Au}_3\text{Fe}_3 - \text{OHCH}_3 \rightarrow \text{Au}_3\text{Fe}_3 + \text{OHCH}_3$ ($\Delta E_{\text{dissociation}}$)	0.67	0.28
$\text{Au}_2\text{Fe}_4 - \text{OHCH}_3 \rightarrow \text{Au}_2\text{Fe}_4 + \text{OHCH}_3$ ($\Delta E_{\text{dissociation}}$)	0.45	0.04
$\text{AuFe}_5 - \text{OHCH}_3 \rightarrow \text{AuFe}_5 + \text{OHCH}_3$ ($\Delta E_{\text{dissociation}}$)	0.63	0.26
$\text{Fe}_6 - \text{OHCH}_3 \rightarrow \text{Fe}_6 + \text{OHCH}_3$ ($\Delta E_{\text{dissociation}}$)	0.54	0.16

In most cases, steps 2 and 4, corresponding to the formation of radicals and the transition states, respectively, are the rate-determining steps of the reaction. The sum of the free energies for these steps ($\Delta G_2 + \Delta G_4$) correlates well with the total free energy of the reaction for each cluster ($\sum_i \Delta G_i$, where i represents the reaction step). This analysis provides insights into which Au/Fe cluster proportion most effectively promotes the spontaneity of the $\text{CH}_4 + \text{H}_2\text{O}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$ reaction,

with Au_3Fe_3 emerging as the most favorable composition. Notably, these clusters drastically reduce the overall free energy barrier compared to the reaction without cluster participation ($398.94 \text{ kcal}\cdot\text{mol}^{-1}$) (Figure S12).

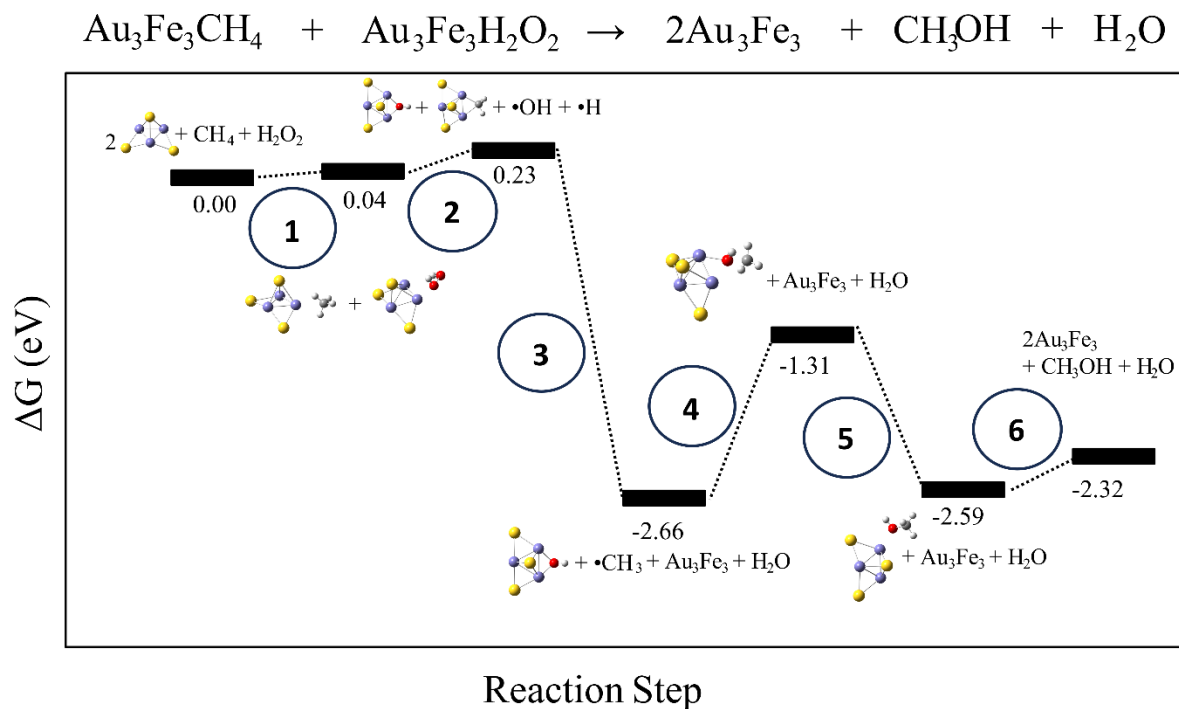


Figure S11. reaction path of the $\text{Au}_3\text{Fe}_3\text{CH}_4 + \text{Au}_3\text{Fe}_3\text{H}_2\text{O}_2 \rightarrow 2\text{Au}_3\text{Fe}_3 + \text{CH}_3\text{OH} + \text{H}_2\text{O}$ reaction.

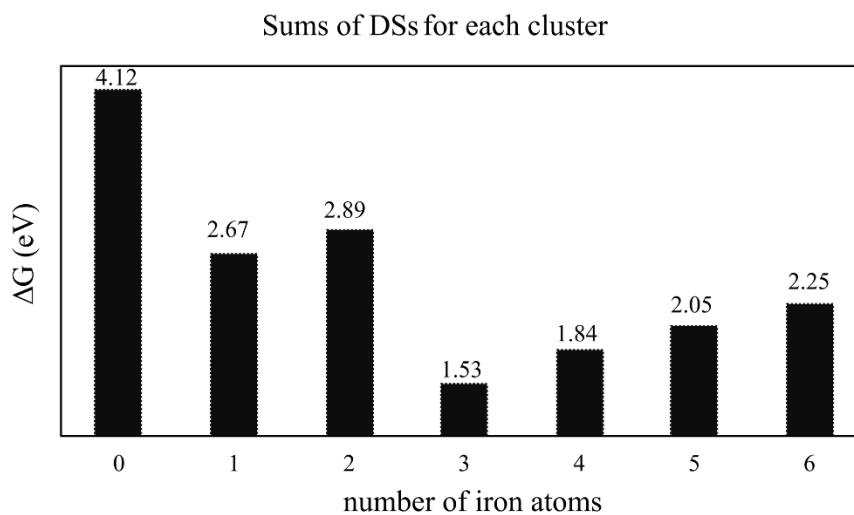


Figure S12. $\sum_i \Delta G_i$ and $\Delta G_2 + \Delta G_4$ reaction paths sums of the determinant steps (DSs), where i is the number of the reaction step, for each Au:Fe cluster proportion.

5. Atomic Coordinates of all systems

Bare clusters

Au ₆			
Au1	1.055373000	1.250284000	0.000448000
Au2	3.032698000	-0.543156000	-0.000408000
Au3	0.555596000	-1.539670000	0.000322000
Au4	-1.611324000	0.289098000	-0.000722000
Au5	-1.986967000	-2.354611000	0.000214000
Au6	-1.045377000	2.898055000	0.000147000
Au ₅ Fe			
Au1	2.256693000	-0.835540000	-0.000801000
Au2	-0.001837000	-2.305674000	-0.000023000
Au3	-2.259625000	-0.833200000	0.000749000
Au4	-2.089339000	1.917161000	-0.000880000
Au5	2.094346000	1.911960000	0.000794000
Fe6	-0.000722000	0.441469000	0.000488000
Au ₄ Fe ₂			
Au1	2.602749000	0.006834000	-0.993896000
Au2	-1.642913000	1.863520000	-0.415775000
Fe3	0.155260000	0.004458000	-0.807386000
Au4	1.077878000	-0.009474000	1.683678000
Au5	-1.640006000	-1.860666000	-0.430433000
Fe6	-1.363683000	-0.005113000	1.282682000
Au ₃ Fe ₃			
Fe1	-0.066179000	-0.395703000	-0.599932000
Fe2	1.150514000	0.384244000	1.462392000
Au3	2.261355000	-1.335470000	-0.133519000
Au4	0.269196000	2.144365000	-0.427294000
Fe5	-1.241221000	0.718729000	1.317643000
Au6	-2.478918000	-1.041667000	-0.156689000
Au ₂ Fe ₄			
Fe1	-0.011060000	0.051529000	1.260624000
Fe2	0.010593000	0.050683000	-1.259912000
Au3	-2.208711000	-0.627583000	-0.000963000
Fe4	1.187574000	1.854807000	0.005823000
Au5	2.208676000	-0.627466000	0.000933000
Fe6	-1.187001000	1.856399000	-0.006443000
AuFe ₅			
Fe1	-2.271236000	-0.001539000	-0.000892000
Fe2	-0.538660000	-0.446954000	1.636855000
Fe3	-0.537738000	0.447316000	-1.636743000
Fe4	-0.582542000	1.746509000	0.419119000
Fe5	-0.579407000	-1.746515000	-0.418974000
Au6	1.484167000	0.000390000	0.000209000

Fe ₆			
Fe1	-0.001428000	-0.001773000	-1.569553000
Fe2	-0.510844000	1.759513000	-0.000753000
Fe3	1.760860000	0.510288000	-0.001920000
Fe4	0.509799000	-1.759215000	0.000867000
Fe5	-1.759734000	-0.509812000	0.001023000
Fe6	0.001346000	0.000999000	1.570336000

Au_xFe_yCH₄

Au₆CH₄

Au1	0.696699000	1.404244000	-0.004403000
Au2	-1.655905000	2.673020000	0.003584000
Au3	-1.775661000	0.006122000	0.003660000
Au4	0.685939000	-1.410237000	-0.004199000
Au5	-1.676138000	-2.661537000	0.003546000
Au6	2.961859000	-0.011430000	-0.012250000
C7	6.025025000	-0.000975000	0.080837000
H8	5.816497000	0.203194000	1.130971000
H9	7.104969000	-0.045891000	-0.067809000
H10	5.598084000	-0.964448000	-0.207476000
H11	5.623603000	0.798572000	-0.545768000

Au₅FeCH₄

Au1	-2.242760000	-0.415224000	-0.000014000
Au2	-0.261250000	-2.243418000	0.000124000
Au3	2.217664000	-1.174020000	-0.000016000
Au4	2.508310000	1.562912000	-0.000120000
Au5	-1.611003000	2.266750000	0.000141000
Fe6	0.198319000	0.457372000	-0.000025000
C7	-5.341738000	-1.164900000	-0.000824000
H8	-4.993406000	-0.129061000	0.004393000
H9	-5.936699000	-1.342938000	0.894165000
H10	-5.943326000	-1.331583000	-0.893578000
H11	-4.498336000	-1.861678000	-0.008421000

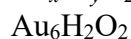
Au₄Fe₂CH₄

Au1	2.583984000	0.002165000	0.944763000
Au2	-1.638410000	-1.876857000	0.255548000
Fe3	0.122704000	0.001700000	0.753086000
Au4	1.153042000	-0.003762000	-1.711921000
Au5	-1.638498000	1.877988000	0.248351000
Fe6	-1.297633000	-0.002708000	-1.425324000
C7	-0.580153000	0.006328000	3.826732000
H8	-0.706418000	-0.884624000	4.441031000
H9	0.429288000	0.003881000	3.404948000
H10	-1.336972000	0.002271000	3.037314000
H11	-0.706198000	0.903460000	4.431970000

Au₃Fe₃CH₄

Fe1	0.132170000	-0.561517000	-0.558497000
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Fe2	0.989017000	0.526592000	1.524604000
Au3	2.545333000	-1.021852000	0.104192000
Au4	0.217555000	2.009101000	-0.678926000
Fe5	-1.391186000	0.671408000	1.003317000
Au6	-2.225224000	-1.453814000	-0.295860000
C7	-3.538879000	2.027614000	1.760918000
H8	-3.108902000	2.462067000	2.662196000
H9	-4.530973000	2.443711000	1.583829000
H10	-2.929631000	2.294129000	0.883502000
H11	-3.652701000	0.944553000	1.876857000
Au ₂ Fe ₄ CH ₄			
Fe1	0.116557000	-0.055565000	1.273900000
Fe2	0.124806000	-0.017599000	-1.245677000
Au3	-1.909921000	-1.137386000	-0.003797000
Fe4	0.969124000	1.938756000	0.032326000
Au5	2.418971000	-0.336936000	-0.015595000
Fe6	-1.400155000	1.516192000	0.029324000
C7	-3.519143000	2.852750000	-0.082438000
H8	-2.955493000	3.600797000	-0.642006000
H9	-3.110810000	2.741158000	0.932321000
H10	-4.551030000	3.187133000	0.028199000
H11	-3.551356000	1.899511000	-0.628615000
AuFe ₅ CH ₄			
Fe1	-2.072463000	0.001289000	0.007140000
Fe2	-0.316695000	0.358552000	-1.632872000
Fe3	-0.308010000	-0.356808000	1.637288000
Fe4	-0.362941000	-1.772170000	-0.326948000
Fe5	-0.359422000	1.772600000	0.333018000
Au6	1.724932000	-0.000822000	-0.004096000
C7	-4.728882000	-0.002834000	-0.014028000
H8	-5.819228000	0.021714000	-0.005056000
H9	-4.411053000	-0.897865000	-0.556059000
H10	-4.373997000	0.903821000	-0.513894000
H11	-4.384292000	-0.035801000	1.024482000
Fe ₆ CH ₄			
Fe1	0.281419000	-0.666264000	-1.570685000
Fe2	2.048984000	-0.005172000	-0.002273000
Fe3	0.282713000	-1.668102000	0.627009000
Fe4	-1.469581000	0.005033000	0.002558000
Fe5	0.291865000	1.667950000	-0.627544000
Fe6	0.290406000	0.665863000	1.570956000
C7	-4.483416000	0.001945000	0.000099000
H8	-5.572814000	-0.029099000	-0.042820000
H9	-4.171374000	-0.005145000	1.046125000
H10	-4.087991000	-0.875405000	-0.516925000
H11	-4.138297000	0.915963000	-0.487550000



Au	-0.680263000	1.337955000	-0.026907000
Au	1.596059000	2.751715000	0.013369000
Au	1.883550000	0.098552000	0.022386000
Au	-0.517392000	-1.435662000	-0.020786000
Au	1.911499000	-2.569166000	0.027690000
Au	-2.878371000	-0.186684000	-0.064605000
O	-5.367528000	-0.283976000	-0.292510000
H	-5.659330000	0.233496000	-1.055267000
O	-6.082512000	0.337110000	0.772562000
H	-6.631791000	-0.398621000	1.074211000



Au1	2.283746000	-0.718348000	-0.121120000
Au2	0.114070000	-2.309012000	-0.116127000
Au3	-2.181484000	-0.896455000	-0.243329000
Au4	-2.231485000	1.830878000	-0.043649000
Au5	2.033759000	2.003165000	-0.154360000
Fe6	-0.012603000	0.489441000	0.230923000
O7	-0.251719000	0.311748000	2.356478000
H8	-1.201172000	0.310214000	2.551707000
O9	0.217509000	-0.871027000	2.996463000
H10	0.332674000	-1.469409000	2.229024000



Au	-2.585579000	0.208009000	0.888937000
Au	1.199187000	1.947632000	-0.741579000
Fe	-0.084225000	0.156287000	0.771088000
Au	-1.129671000	-0.912227000	-1.440475000
Au	1.945338000	-1.512333000	0.892620000
Fe	1.306707000	-0.531936000	-1.357116000
O	0.255941000	1.198243000	2.627232000
H	-0.554822000	1.705277000	2.796344000
O	1.286057000	2.183578000	2.667958000
H	1.521648000	2.251625000	1.718150000



Fe1	0.169374000	-0.744363000	-0.514888000
Fe2	0.887886000	0.678241000	1.422684000
Au3	2.569896000	-0.969751000	0.267948000
Au4	0.234894000	1.776957000	-1.054302000
Fe5	-1.487365000	0.699961000	0.716290000
Au6	-2.180573000	-1.638881000	-0.187630000
O7	-2.457635000	2.521711000	1.336792000
H8	-2.187482000	3.235375000	0.739498000
O9	-1.904550000	2.922083000	2.588280000
H10	-1.045534000	2.436754000	2.578346000



Fe1	0.171745000	-0.965645000	-1.135838000
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Fe2	-0.023951000	0.477638000	0.947571000
Au3	-2.209164000	0.150807000	-0.471274000
Fe4	1.342655000	-1.619050000	1.004297000
Au5	2.260161000	0.493379000	-0.381041000
Fe6	-1.011847000	-1.827520000	0.895557000
H7	0.246062000	3.160566000	0.777534000
O8	-0.253739000	2.627405000	1.412381000
O9	-1.586797000	3.108282000	1.249963000
H10	-1.994201000	2.362284000	0.755289000
AuFe ₅ H ₂ O ₂			
Fe1	-1.904645000	0.591564000	0.279065000
Fe2	-0.539205000	-1.422431000	0.891115000
Fe3	0.057842000	1.762289000	-0.729377000
Fe4	0.159023000	0.896168000	1.528691000
Fe5	-0.501553000	-0.579544000	-1.348449000
Au6	1.814503000	-0.292143000	-0.133587000
O7	-4.001245000	0.184544000	-0.183655000
H8	-4.246649000	0.505449000	-1.062854000
O9	-4.122248000	-1.232288000	-0.335541000
H10	-3.169144000	-1.493419000	-0.377330000
Fe ₆ H ₂ O ₂			
Fe1	-2.029189000	0.761878000	-0.071137000
Fe2	0.160466000	1.296036000	1.061095000
Fe3	0.118054000	0.965739000	-1.298665000
Fe4	-0.985906000	-1.297333000	-1.056723000
Fe5	-0.917001000	-0.943682000	1.314900000
Fe6	1.198650000	-0.776395000	0.071050000
O7	3.412724000	-0.680644000	0.180714000
H8	3.742089000	-0.623258000	1.088509000
O9	3.742684000	0.610005000	-0.337826000
H10	2.842730000	1.026037000	-0.365163000
Au _x Fe _y CH ₃ •			
Au ₆ CH ₃ •			
Au1	0.771750000	1.364185000	0.000085000
Au2	-1.553048000	2.727140000	-0.000015000
Au3	-1.726527000	0.068170000	-0.000010000
Au4	0.653374000	-1.437931000	0.000094000
Au5	-1.788590000	-2.594532000	-0.000014000
Au6	3.049074000	-0.120984000	-0.000253000
C7	5.111393000	-0.054387000	0.000946000
H8	5.408386000	0.467663000	0.909295000
H9	5.409019000	0.475300000	-0.902768000
H10	5.437690000	-1.094500000	-0.003295000
Au ₅ FeCH ₃ •			
Au1	-2.548341000	-0.286336000	0.000178000
Au2	-0.359731000	-2.052810000	-0.000153000

Au3	2.134949000	-1.260028000	-0.000082000
Au4	2.673062000	1.433463000	0.000178000
Au5	-1.580118000	2.242494000	-0.000215000
Fe6	0.259627000	0.531868000	0.000073000
C7	-3.490116000	-2.112574000	0.000595000
H8	-4.551214000	-1.844456000	0.000701000
H9	-3.261969000	-2.687133000	0.898038000
H10	-3.262259000	-2.687438000	-0.896729000
Au ₄ Fe ₂ CH ₃ •			
Au1	-2.590511000	0.002502000	0.582728000
Au2	1.430087000	2.113089000	0.274160000
Fe3	-0.043141000	0.001439000	1.050098000
Au4	-0.820295000	-0.003105000	-1.506323000
Au5	1.427986000	-2.113016000	0.278164000
Fe6	1.667411000	-0.001208000	-1.013706000
C7	0.149937000	0.003784000	3.036882000
H8	-0.336592000	0.899100000	3.429525000
H9	-0.337313000	-0.889896000	3.432221000
H10	1.209201000	0.003902000	3.301229000
Au ₃ Fe ₃ CH ₃ •			
Fe1	-0.053032000	-0.454395000	-0.696337000
Fe2	1.083370000	0.371842000	1.373867000
Au3	2.286213000	-1.352262000	-0.157216000
Au4	0.271646000	2.114567000	-0.607157000
Fe5	-1.198422000	0.727279000	1.148179000
Au6	-2.493221000	-1.071255000	-0.201964000
C7	-0.053285000	0.818616000	3.070036000
H8	0.577865000	1.631567000	3.450733000
H9	0.084065000	-0.067002000	3.698670000
H10	-1.078370000	1.167920000	3.302509000
Au ₂ Fe ₄ CH ₃ •			
Fe1	0.103268000	0.056643000	1.300916000
Fe2	0.103032000	0.055756000	-1.300899000
Au3	-1.776070000	-1.196705000	0.000004000
Fe4	0.903588000	2.000286000	-0.000853000
Au5	2.351020000	-0.330019000	0.000192000
Fe6	-1.691737000	1.528932000	-0.000056000
C7	-3.264351000	2.796394000	0.000804000
H8	-2.914211000	3.834929000	0.000685000
H9	-3.895930000	2.657953000	0.885345000
H10	-3.896747000	2.657971000	-0.883153000
AuFe ₅ CH ₃ •			
Fe1	-2.157427000	0.681211000	-0.143398000
Fe2	-0.396548000	-0.520249000	1.291441000
Fe3	-0.426931000	-0.584436000	-1.288106000
Fe4	-0.064133000	1.581301000	-0.040717000
Fe5	-2.072069000	-1.777281000	0.093948000

Au6	1.890320000	-0.130102000	-0.000134000
C7	-1.735813000	2.812230000	0.222300000
H8	-1.654893000	3.530079000	-0.600319000
H9	-1.394004000	3.289110000	1.149276000
H10	-2.826696000	2.691274000	0.385440000

Fe₆CH₃•

Fe1	-0.261453000	-0.467736000	1.674128000
Fe2	-2.021526000	-0.006732000	-0.003072000
Fe3	-0.257872000	-1.670641000	-0.469749000
Fe4	1.724665000	0.004996000	0.001799000
Fe5	-0.267593000	1.672750000	0.466454000
Fe6	-0.261655000	0.468733000	-1.668391000
C7	3.752682000	-0.003201000	-0.003062000
H8	4.160988000	0.929657000	0.404292000
H9	4.152601000	-0.119384000	-1.017731000
H10	4.151585000	-0.826713000	0.601427000

Au_xFe_yHO•

Au₆HO•

Au1	-0.726928000	-1.428144000	0.049561000
Au2	1.660258000	-2.663488000	-0.012262000
Au3	1.688092000	0.000552000	-0.009289000
Au4	-0.728038000	1.427669000	0.049647000
Au5	1.658053000	2.664683000	-0.012306000
Au6	-2.985060000	-0.001070000	-0.097828000
O7	-4.951252000	-0.001703000	0.179887000
H8	-5.133825000	-0.002346000	1.126601000

Au₅FeHO•

Au1	-0.753709000	2.199329000	-0.160798000
Au2	1.881495000	1.392670000	-0.148221000
Au3	1.894740000	-1.363901000	-0.140939000
Au4	-0.698899000	-2.206754000	-0.148577000
Au5	-2.328489000	-0.035505000	-0.138376000
Fe6	0.021620000	0.019402000	1.186515000
O7	0.033176000	-0.001515000	2.980387000
H8	-0.443407000	0.626404000	3.523550000

Au₄Fe₂HO•

Au1	-1.435891000	1.576343000	-0.059388000
Au2	1.895167000	-1.466712000	-0.149701000
Fe3	-0.000029000	-0.262069000	1.384854000
Au4	1.435726000	1.576363000	-0.059079000
Au5	-1.894959000	-1.466981000	-0.149542000
Fe6	-0.000001000	-0.203783000	-1.228225000
O7	-0.000217000	-0.480396000	3.168716000
H8	-0.000861000	-1.346697000	3.577057000

Au₃Fe₃HO•

Fe1	0.030208000	-0.562720000	-0.613959000
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Fe2	1.133911000	0.476717000	1.274074000
Au3	2.528438000	-1.135928000	-0.158772000
Au4	-0.011300000	2.111480000	-0.566894000
Fe5	-1.319332000	0.619159000	1.171065000
Au6	-2.447909000	-1.227527000	-0.198351000
O7	-0.156668000	0.601449000	2.740707000
H8	-0.180250000	1.232402000	3.461055000
Au ₂ Fe ₄ HO•			
Fe1	-0.005284000	-0.164949000	-1.282029000
Fe2	0.017677000	0.224532000	1.263393000
Au3	2.220885000	-0.647444000	-0.039134000
Fe4	-1.201173000	1.800839000	-0.368115000
Au5	-2.227005000	-0.645358000	-0.035319000
Fe6	1.183730000	1.802562000	-0.439881000
H7	0.320901000	1.543630000	3.278942000
O8	0.036731000	0.668764000	3.011914000
AuFe ₅ HO•			
Fe1	2.327446000	0.016670000	0.009590000
Fe2	0.391242000	-1.702129000	0.224155000
Fe3	0.345971000	1.723539000	-0.223271000
Fe4	0.381849000	-0.219431000	-1.698758000
Fe5	0.353310000	0.224657000	1.713278000
Au6	-1.729330000	-0.012792000	-0.006814000
O7	4.137547000	0.061169000	0.027713000
H8	4.721442000	-0.604760000	-0.333214000
(Au _x Fe _y OH-CH ₃) _{ts}			
(Au ₆ OH-CH ₃) _{ts}			
Au1	0.597005000	-1.381977000	-0.058485000
Au2	-1.793751000	-2.614915000	0.038678000
Au3	-1.890868000	0.048780000	0.041292000
Au4	0.669652000	1.327688000	-0.063292000
Au5	-1.633876000	2.706759000	0.023376000
Au6	2.979619000	-0.099370000	-0.157852000
O7	5.090264000	-0.118581000	-0.179265000
H8	5.330457000	0.784199000	-0.422707000
C9	4.279232000	0.124708000	1.662936000
H10	4.767617000	-0.807831000	1.904314000
H11	4.904617000	1.000913000	1.758725000
H12	3.305094000	0.252788000	2.142632000
(Au ₅ FeOH-CH ₃) _{ts}			
Au1	-0.915871000	1.702043000	-0.593934000
Au2	1.740778000	1.484117000	-0.289978000
Au3	2.060320000	-1.316281000	-0.019135000
Au4	-0.530479000	-1.720884000	-0.671870000
Au5	-2.554022000	-0.254380000	0.404688000
Fe6	0.078794000	0.014529000	1.216801000

O7	-0.206560000	0.117184000	3.043157000
H8	-0.708577000	0.870823000	3.366540000
C9	1.739518000	0.672269000	3.658471000
H10	1.791398000	1.709159000	3.357226000
H11	1.488242000	0.438758000	4.681145000
H12	2.338297000	-0.042135000	3.110217000
(Au ₄ Fe ₂ OH-CH ₃) _{ts}			
Au	-2.316810000	-0.018767000	-0.461022000
Au	2.136435000	0.020458000	0.247229000
Fe	-0.301270000	-0.005403000	1.219555000
Au	0.261830000	2.271140000	-0.295998000
Au	0.304929000	-2.266536000	-0.300569000
Fe	0.067152000	0.001238000	-1.366872000
O	-0.581648000	-0.011316000	3.070623000
H	0.204510000	-0.004968000	3.628751000
C	-2.186208000	-0.032168000	4.380285000
H	-1.972426000	0.898428000	4.883821000
H	-2.944849000	-0.036797000	3.612593000
H	-1.954030000	-0.962085000	4.876878000
(Au ₃ Fe ₃ OH-CH ₃) _{ts}			
Fe1	-0.009012000	-0.345875000	-0.864575000
Fe2	1.140463000	0.390610000	1.229790000
Au3	2.301654000	-1.330371000	-0.299827000
Au4	0.209587000	2.211458000	-0.584642000
Fe5	-1.252537000	0.653163000	1.058275000
Au6	-2.415099000	-1.121475000	-0.416479000
O7	-0.155101000	0.587497000	2.857097000
H8	-0.066794000	1.478222000	3.239971000
C9	-0.341706000	-0.549922000	4.356443000
H10	-0.448685000	-1.487261000	3.829429000
H11	0.592883000	-0.392193000	4.876868000
H12	-1.233359000	-0.153872000	4.822530000
(Au ₂ Fe ₄ OH-CH ₃) _{ts}			
Fe1	0.009242000	-1.491045000	-0.110709000
Fe2	0.009245000	1.076231000	0.190889000
Au3	-2.207132000	-0.229722000	-0.642586000
Fe4	1.200375000	-0.515545000	1.820456000
Au5	2.206645000	-0.192300000	-0.652872000
Fe6	-1.178463000	-0.493857000	1.811643000
H7	0.191471000	3.054237000	1.556620000
O8	0.020291000	2.918672000	0.616741000
C9	-0.146985000	4.839120000	-0.055376000
H10	0.853789000	5.183032000	0.160367000
H11	-0.381937000	4.575388000	-1.074965000
H12	-0.955673000	5.172625000	0.578168000
(AuFe ₅ OH-CH ₃) _{ts}			
Fe1	-2.018662000	0.135198000	0.052499000

Fe2	-0.113938000	1.783747000	-0.020311000
Fe3	-0.232574000	-1.759469000	0.054648000
Fe4	-0.166984000	-0.011946000	-1.667841000
Fe5	-0.090391000	0.044066000	1.702071000
Au6	1.938772000	-0.057303000	-0.031105000
O7	-3.859385000	0.520109000	0.091265000
H8	-4.184899000	1.332457000	-0.301106000
C9	-5.500228000	-0.625184000	-0.120730000
H10	-5.114957000	-1.482857000	0.409667000
H11	-6.233580000	-0.019038000	0.392638000
H12	-5.566843000	-0.694889000	-1.197333000

(Fe₆OH-CH₃)_{ts}

Fe1	2.132308000	0.004537000	-0.001171000
Fe2	0.506569000	1.294305000	1.210130000
Fe3	0.530575000	-1.199632000	1.296343000
Fe4	0.533087000	-1.254867000	-1.217584000
Fe5	0.512746000	1.228013000	-1.273102000
Fe6	-1.310920000	-0.007826000	-0.004144000
O7	-3.129011000	-0.401287000	-0.067435000
H8	-3.309416000	-1.334448000	-0.224085000
C9	-5.210636000	0.305939000	0.052479000
H10	-4.911227000	1.328619000	0.217408000
H11	-5.500646000	0.000545000	-0.941334000
H12	-5.496332000	-0.297820000	0.900340000

Au_xFe_y-OHCH₃

Au₆-OHCH₃

Au1	-0.760157000	1.273854000	-0.040572000
Au2	1.420610000	2.831222000	0.011520000
Au3	1.875938000	0.199295000	0.030134000
Au4	-0.437608000	-1.475041000	-0.026275000
Au5	2.061850000	-2.458522000	0.043455000
Au6	-2.869082000	-0.385847000	-0.095081000
O7	-5.277907000	-0.318095000	-0.185916000
H8	-5.547537000	0.045737000	-1.033765000
C9	-5.971815000	0.356715000	0.860194000
H10	-5.655696000	-0.103319000	1.794433000
H11	-5.723697000	1.420851000	0.885904000
H12	-7.051476000	0.229365000	0.748286000

Au₅Fe-OHCH₃

Au	-2.225225000	1.823532000	-0.117542000
Au	2.035088000	1.985882000	-0.151434000
Au	2.281631000	-0.739921000	-0.178558000
Au	0.102877000	-2.314909000	-0.089863000
Au	-2.187052000	-0.906338000	-0.204958000
Fe	-0.008643000	0.461640000	0.270884000
O	-0.239149000	0.534168000	2.380086000

H	-1.168334000	0.716903000	2.557740000
C	0.255364000	-0.472489000	3.262679000
H	1.306442000	-0.610482000	3.016825000
H	0.162528000	-0.142210000	4.299440000
H	-0.273101000	-1.416654000	3.112281000
Au ₄ Fe ₂ —OHCH ₃			
Au1	-1.016531000	-1.448298000	-1.578218000
Au2	3.160582000	0.226335000	-0.392146000
Fe3	-1.783793000	-0.095520000	0.510893000
Au4	-0.805317000	2.171495000	-0.063701000
Au5	-0.019266000	-1.038052000	1.985224000
Fe6	0.740160000	0.035220000	-0.344568000
O7	-3.782065000	-0.087466000	-0.109087000
H8	-3.773532000	-0.321115000	-1.045438000
C9	-4.709067000	0.977303000	0.134782000
H10	-5.709446000	0.674288000	-0.177062000
H11	-4.707495000	1.158183000	1.207866000
H12	-4.402197000	1.885413000	-0.387305000
Au ₃ Fe ₃ —OHCH ₃			
Fe1	-0.297051000	-0.641747000	-0.525576000
Fe2	1.433351000	0.481055000	0.931595000
Au3	1.942692000	-1.814803000	-0.284257000
Au4	-0.077671000	1.915572000	-0.752068000
Fe5	-0.934798000	0.583080000	1.554132000
Au6	-2.746368000	-0.696735000	0.144086000
C7	3.692713000	2.651439000	0.922951000
H8	4.613274000	2.966286000	1.417997000
H9	2.870419000	3.283111000	1.252651000
H10	3.793364000	2.733222000	-0.160932000
O11	3.362273000	1.312331000	1.306757000
H12	4.055832000	0.709315000	1.021447000
Au ₂ Fe ₄ —OHCH ₃			
Fe1	0.211847000	-1.557091000	-0.026440000
Fe2	-0.022367000	0.980868000	0.036766000
Au3	-2.184180000	-0.386448000	-0.609386000
Fe4	1.256591000	-0.212750000	1.838091000
Au5	2.303460000	0.004333000	-0.617429000
Fe6	-1.089830000	-0.452449000	1.811103000
O7	-0.977146000	2.883179000	-0.375573000
H8	-1.877098000	2.581952000	-0.562588000
C9	-1.003370000	3.995315000	0.518048000
H10	-1.474487000	3.730028000	1.467639000
H11	0.031641000	4.277974000	0.701490000
H12	-1.528019000	4.836804000	0.060628000
AuFe ₅ —OHCH ₃			
Fe1	-1.941747000	0.169728000	0.200429000
Fe2	-0.118315000	1.791185000	-0.409039000

Fe3	-0.282131000	-1.696441000	0.531108000
Fe4	-0.300583000	-0.420760000	-1.513535000
Fe5	0.013706000	0.510377000	1.628333000
Au6	1.901266000	-0.093783000	-0.104757000
O7	-4.062050000	0.396371000	0.133923000
H8	-4.332434000	1.267911000	-0.170928000
C9	-4.931898000	-0.604301000	-0.402457000
H10	-4.566247000	-1.562805000	-0.039818000
H11	-5.951443000	-0.445774000	-0.045780000
H12	-4.906288000	-0.601943000	-1.494003000
Fe ₆ -OHCH ₃			
Fe1	-2.223903000	0.234537000	0.177017000
Fe2	-0.731558000	-1.667392000	-0.517424000
Fe3	-0.369851000	-0.570708000	1.588467000
Fe4	-0.190508000	1.681637000	0.516210000
Fe5	-0.550798000	0.583979000	-1.588841000
Fe6	1.320577000	-0.222992000	-0.184868000
O7	3.433595000	-0.573257000	-0.139342000
H8	3.647203000	-1.353391000	0.382981000
C9	4.397499000	0.456664000	0.093708000
H10	5.385353000	0.129140000	-0.236276000
H11	4.085757000	1.315685000	-0.497016000
H12	4.424979000	0.739039000	1.148241000

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